

Article

Enhancing CIA Triad—Confidentiality, Integrity and Availability in Educational Information Systems Through Next-Generation ISO/IEC 27001:2022-Aligned Security Model

Dejan Vasović ^{1,*} , Goran Janačković ¹ , Žarko Vranjanac ² , Srećko Stamenković ³  and Bojan Vasović ³ 

¹ Faculty of Occupational Safety, University of Niš, Čarnojevića 10a, 18000 Niš, Serbia; goran.janackovic@znrfak.ni.ac.rs

² Innovation Center, University of Niš, Univerzitetski trg 2, 18000 Niš, Serbia; zarko.vranjanac@icun.ni.ac.rs

³ Toplica Academy of Applied Studies, Department of Business Studies, Kralja Petra I 70, 18420 Blace, Serbia; stamenkovic.srecko@gmail.com (S.S.); bojan.vasovic@yahoo.com (B.V.)

* Correspondence: djnvasovic@gmail.com or dejan.vasovic@znrfak.ni.ac.rs; Tel.: +381-65-5-883-883

Featured Application

This study can be applied to strengthening the security architecture of digital education platforms by integrating next-generation security protocols into ISO/IEC 27001-aligned governance models. It provides guidance elements for educational institutions seeking to enhance the CIA triad in increasingly interconnected learning environments.

Abstract

Educational information systems have evolved into highly interconnected digital landscapes that support learning management platforms, student information systems, institutional repositories, and online assessment environments. As these systems increasingly operate across cloud infrastructures and mobile devices, ensuring the confidentiality, integrity, and availability (CIA Triad) of educational data is critical for safeguarding institutional operations and maintaining trust in digital education services. This paper investigates how next-generation security protocols, such as adaptive multi-factor authentication and advanced access control and data protection mechanisms, can reinforce ISO/IEC 27001:2022 requirements within contemporary educational information systems. The analysis maps emerging protocol capabilities to relevant new ISO/IEC 27001:2022 control domains, illustrating how they mitigate threats associated with unauthorized access, data manipulation, and service disruption. The proposed framework is further supported by an implementation-oriented mapping and an illustrative operational architecture that demonstrates the feasibility of translating prioritized security determinants into practical mechanisms. The FAHP analysis identifies access control mechanisms, backup and recovery, and data validation as the three highest-weighted determinants, with aggregate weights of 0.061, 0.059, and 0.057, respectively. These determinants are translated into a determinant-driven Security Operationalization Matrix that connects ISO/IEC 27001:2022 control domains, CIA dimensions, and technology recommendations, and is complemented by implementation feasibility considerations tailored to the budgetary, infrastructural, and resource constraints characteristic of educational institutions. Based on the prioritization results and conceptual operationalization, the proposed integrative approach provides a structured and progressively adoptable foundation for CIA-oriented security governance in digital educational environments.

Keywords: ISO/IEC 27001; information security; confidentiality; integrity; availability; educational information systems; digital threats; secure-by-design; model



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1. Introduction

In the context of accelerating digital transformation, educational information systems (EIS) have become essential infrastructures supporting modern education and institutional governance. These systems process substantial volumes of personal, academic, and operational data, enabling digital learning, analytics, and evidence-based decision-making [1–5]. The rapid growth of cloud computing, ubiquitous connectivity, and data-driven educational environments has increased both the strategic value of information and the exposure of educational institutions to cybersecurity risks [6–9]. Concurrently, cyber threats, including ransomware, phishing, and malware attacks, have become more frequent and sophisticated, particularly in data-intensive organizations [10–16]. Consequently, ensuring the confidentiality, integrity, and availability (CIA) of information is a fundamental requirement for institutional trust, operational continuity, and resilience [17,18]. Contemporary EISs comprise interconnected platforms such as learning management systems, student information systems, digital repositories, and cloud-based educational services [19]. Their increasing integration with mobile technologies, artificial intelligence, and federated digital infrastructures has expanded functionality but also introduced challenges related to privacy, transparency, and regulatory compliance [20–23]. As a result, effective information security management has become critical for safeguarding sensitive educational and operational data [24]. Previous studies have documented the growing cybersecurity challenges facing educational institutions and the uneven adoption of standardized security practices [25]. However, existing research largely concentrates on individual technical controls, threat analyses, or policy perspectives, providing limited insight into the practical implementation of the revised ISO/IEC 27001:2022 framework within educational environments [26–30]. In particular, the relationship between the updated Annex A controls and the CIA triad remains insufficiently explored in educational systems characterized by diverse user populations, federated identity management, and high data turnover [31]. Moreover, existing implementation frameworks rarely account for the resource constraints, legacy infrastructure, high user turnover, and fragmented identity ecosystems that characterize higher education environments, leaving institutions with limited actionable guidance for context-sensitive and phased security adoption. To address these shortcomings, this study moves beyond descriptive analyses of cybersecurity threats and compliance requirements by systematically investigating how ISO/IEC 27001:2022 controls can be operationalized within contemporary educational information systems. Specifically, it examines the contribution of next-generation security protocols to strengthening CIA-oriented protection, identifies and prioritizes key security determinants through a FAHP-based assessment, and develops an implementation-oriented Security Operationalization Matrix that connects ISO/IEC 27001:2022 control domains, CIA objectives, and technology recommendations. In doing so, the study bridges the gap between standard requirements and practical implementation, providing a structured framework tailored to the technological, organizational, and resource constraints of educational institutions.

Accordingly, the main objective of this study is to evaluate how ISO/IEC 27001:2022-aligned security protocols can be adapted to strengthen confidentiality, integrity, and availability within educational information systems while considering institutional and regulatory specificities. In practical terms, the study aims to support educational institutions in identifying which CIA-related controls should be implemented first, how they can be aligned with ISO/IEC 27001:2022, and how they can be progressively deployed under budgetary, technical, and legacy-system constraints. The study proposes a CIA-oriented management model integrating advanced ISO/IEC 27001:2022 controls, governance mechanisms, and multi-criteria decision analysis. The prioritization results indicate that structured alignment between prioritized security controls and institutional informa-

tion management processes can improve strategic coherence in CIA-oriented governance. Rather than directly quantifying reductions in vulnerabilities or improvements in user trust, the proposed framework enables systematic prioritization of security measures according to their relative contribution to CIA dimensions and institutional resilience. Consequently, the study provides a governance-oriented foundation for reducing exposure to dominant threats and strengthening service continuity in educational environments.

The novelty of this research lies in the operational reinterpretation of the CIA triad within EIS through integration with ISO/IEC 27001:2022 control architecture. Unlike previous studies that treat confidentiality, integrity, and availability primarily as normative concepts, the proposed framework translates these dimensions into institutionally applicable security determinants, including access control, data validation, and recovery mechanisms. The resulting model functions simultaneously as a diagnostic tool, implementation roadmap, audit-support mechanism, and strategic decision-support framework for educational institutions. The novelty is therefore not the introduction of standalone cybersecurity technologies, but their prioritized, determinant-driven, and resource-aware integration into an ISO/IEC 27001:2022-aligned EIS security governance model. Through integration of CIA principles with ISO/IEC 27001:2022 controls, the study contributes a conceptual and implementation-oriented decision-support framework supported by illustrative operationalization for resilience-focused information security governance in digital educational ecosystems. Accordingly, the presented operational workflows and architectural mappings serve as illustrative implementation scenarios intended to demonstrate feasibility and methodological coherence. Full-scale institutional deployment and longitudinal quantitative validation remain within the scope of future research.

The following sections examine the role, benefits, and implementation of ISO system standards, with particular emphasis on ISO/IEC 27001:2022. Special attention is given to the identification, evaluation, and prioritization of advanced security determinants relevant to the CIA triad and the development of a secure-by-design management model for educational information systems.

2. Theoretical Background of ISO/IEC 27001:2022 and Related ISO System Standards

ISO/IEC 27001:2022 is the internationally recognized reference standard for establishing, implementing, maintaining, and continually improving an Information Security Management System (ISMS) [32]. The standard adopts a risk-based and process-oriented approach to information security, emphasizing the systematic identification, assessment, treatment, and monitoring of threats that may compromise organizational information assets. Its conceptual foundation is rooted in the Confidentiality-Integrity-Availability (CIA) triad, which defines the core security objectives that guide both strategic governance and operational control implementation. Within this framework, confidentiality protects information from unauthorized disclosure, integrity ensures the accuracy and reliability of information throughout its lifecycle, and availability guarantees timely and authorized access to information and services. Structurally, ISO/IEC 27001:2022 follows the Harmonized Structure (HS), formerly Annex SL, which standardizes the core management system requirements contained in Clauses 4–10 and incorporates climate change considerations introduced through ISO amendments in 2024 [33]. These clauses address organizational context, leadership, planning, support, operation, performance evaluation, and continual improvement. The adoption of the Harmonized Structure enables direct integration with other ISO management system standards by aligning terminology, governance processes, documented information requirements, risk-based thinking, and

performance evaluation mechanisms. Consequently, information security management can be embedded within broader organizational management systems rather than operating as an isolated compliance function. The integration between ISO/IEC 27001:2022 and ISO 9001:2015 is particularly evident through their shared emphasis on process management, evidence-based decision-making, risk-based thinking, and continual improvement [34]. In this relationship, information security supports quality objectives by protecting the accuracy, reliability, and availability of information required for consistent service delivery, stakeholder satisfaction, and organizational learning. Similarly, ISO/IEC 27001:2022 complements ISO 14001:2015 through the protection of environmental information, monitoring data, compliance records, and sustainability reporting processes [35]. Secure management of environmental information enhances data credibility, regulatory compliance, and the transparency of environmental management practices. A comparable synergy exists with ISO 45001:2018, where information security contributes to occupational health and safety management by ensuring the confidentiality and integrity of employee records, incident reports, risk assessments, and emergency response information [36]. This integration highlights the human-centric dimension of information security, recognizing that organizational resilience depends not only on technological safeguards but also on the protection of information supporting employee well-being and operational safety. While ISO/IEC 27001:2022 establishes the requirements for an ISMS, its practical implementation is strengthened through integration with complementary standards from the ISO/IEC 27000 family. ISO/IEC 27002:2022 provides detailed guidance for the selection, implementation, and management of information security controls, thereby translating ISMS requirements into operational practices. ISO/IEC 27005:2022 further reinforces the framework through structured approaches to information security risk identification, analysis, evaluation, and treatment, ensuring alignment between security controls and organizational risk exposure [37,38]. In addition, ISO 19011:2018 provides common principles and methodologies for auditing management systems, facilitating consistent performance evaluation, compliance verification, and continual improvement across integrated management system environments [39]. Collectively, these standards form a coherent conceptual framework in which information security is integrated with quality, environmental, and occupational health and safety management, supporting a holistic approach to organizational governance and resilience.

The 2022 revision of ISO 27001 system standard further strengthens the alignment with contemporary digital ecosystems by emphasizing organizational context and dynamic threat treatment (Figure 1a), while modernizing Annex A controls (Figure 1b) reflects current technological, organizational, physical and human security challenges.

In the context of educational information systems, the integration of ISO/IEC 27001:2022 with other harmonized management system standards acquires particular significance, as educational institutions increasingly rely on complex digital infrastructures to support teaching, learning, research, and administrative functions. The broader ISO/IEC 27000 family further strengthens this framework by offering control guidance, and privacy-oriented extensions that address the specific regulatory, ethical, and technological challenges faced by contemporary educational institutions. Collectively, these interconnections position information security management not merely as a technical safeguard, but as an integral component of educational governance and performance, thereby providing a robust theoretical foundation for the analytical and methodological approaches applied in this study.

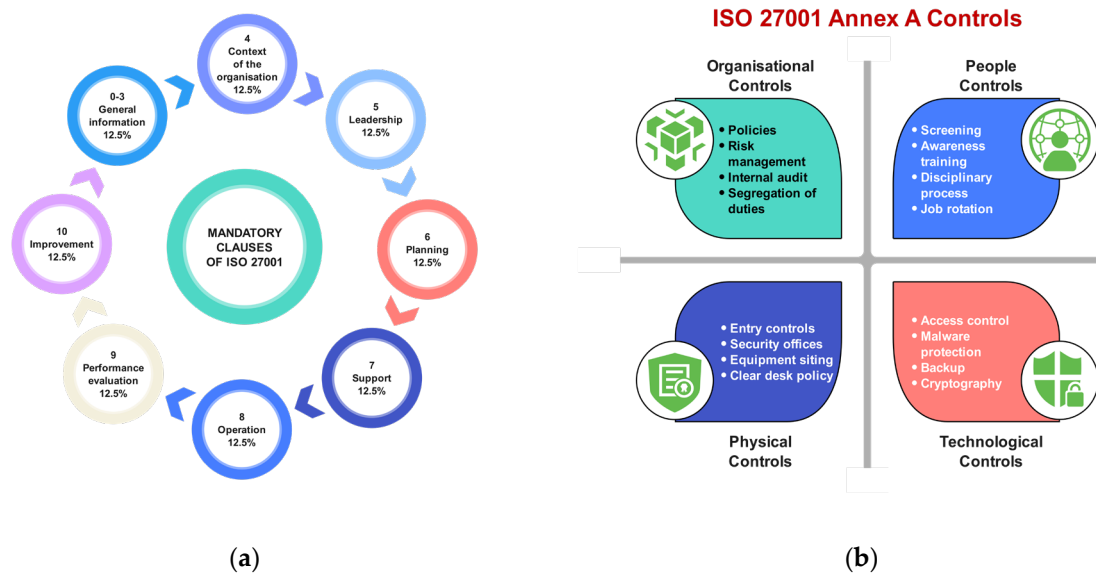


Figure 1. New features under the ISO/IEC 27001:2022: (a) Harmonized structure clauses; (b) New ISMS controls provided in Annex A. Note: Source: authors.

3. Materials and Methods

The implemented research framework adopts a three-layer architecture: brief ISO 27001:2022 trend analysis, MCDA-based ISMS determinant prioritisation and CIA information security model implementation study. Such unified research framework systematically integrates all analytical components into a coherent and logical scientific structure. The framework is designed to support primary research objective: the operational enhancement of the Confidentiality–Integrity–Availability (CIA) triad management model within Educational Information Systems (EIS) through ISO/IEC 27001:2022-aligned information security mechanisms.

At the contextual level, the ISO certification trend analysis provides basic empirical grounding by positioning ISO/IEC 27001:2022 within global standardization dynamics and demonstrating its increasing relevance in digitally intensive environments. At the analytical level, the study employs a Multi-Criteria Decision Analysis (MCDA) approach based on the fuzzy Analytic Hierarchy Process (FAHP). Within this layer, the CIA triad is transformed from a conceptual security paradigm into a measurable decision framework composed of criteria, factors, and 25 determinants. This transformation enables the systematic prioritization of security elements in accordance with institutional constraints. At the operational level, prioritized determinants are translated into concrete technical solutions and implementation mechanisms. This layer ensures that analytical outputs are directly applicable within real-world EIS environments.

To ensure structural coherence, the research workflow is explicitly defined as a sequential yet iterative process:

- Empirical contextualization through ISO trend analysis;
- Conceptual structuring via CIA-ISO alignment;
- Analytical prioritization using FAHP-based MCDA;
- Operational translation into proposed controls and technologies;
- Illustrative operationalization and refinement.

This workflow directly reflects the Plan-Do-Check-Act (PDCA) logic embedded in ISO/IEC 27001:2022, enabling continuous feedback between implementation and evaluation phases. By explicitly linking all methodological components, the framework establishes a coherent, end-to-end research design.

3.1. ISO System Standard Certifications—ISO Survey Role

The ISO Survey of Management System Standard Certifications constitutes a methodologically robust empirical source for conducting longitudinal and comparative trend analyses of ISO/IEC 27001:2022 adoption [40]. Unlike general ICT indicator databases, the ISO Survey systematically records the annual number of valid certificates and sites across countries, regions, and economic sectors using a harmonized and internationally standardized data-collection procedure. In the context of this study, the database was explicitly used to define very basic trends in ISO/IEC 27001 certifications and interpret shifts in institutional uptake in relation to evolving information security threats and governance demands. For research focused on ISO/IEC 27001:2022, the ISO Survey provides a direct and outcome-based proxy for institutional commitment to ISO principles, as it reflects verified certification achievements rather than indirect sustainable capacity indicators [41].

3.2. Application of MCDA to Select the Most Influential CIA Determinants in ISO/IEC 27001:2022

ISO/IEC 27001:2022 provides a comprehensive framework for establishing, implementing, maintaining, and continually improving an Information Security Management System (ISMS). By aligning security objectives with organizational requirements, the standard supports structured and auditable protection of information assets while ensuring the confidentiality, integrity, and availability (CIA) of information throughout its lifecycle, as defined in ISO 14072:2024 [42]. Nevertheless, the complexity of ISO/IEC 27001:2022 presents significant implementation challenges. The standard incorporates numerous mandatory requirements and Annex A controls spanning technical, organizational, human, and physical security dimensions. Through the embedded Plan-Do-Check-Act (PDCA) cycle, organizations must continuously monitor, evaluate, and adapt security controls in response to evolving threats, technological changes, and organizational priorities [43]. Consequently, information security evolves from a static compliance activity into a dynamic governance function based on continuous improvement. However, within resource-constrained and heterogeneous environments such as higher education institutions, equal prioritization of all CIA-related activities is rarely feasible. This limitation emphasizes the need for adaptive prioritization mechanisms capable of identifying critical controls, allocating resources efficiently, and supporting resilient governance models. In this regard, integrating the ISO/IEC 27001:2022 PDCA cycle with analytical and expert-based prioritization approaches enhances institutional capacity to manage uncertainty and strengthen CIA-oriented protection mechanisms. Accordingly, effective ISMS implementation requires systematic prioritization of security objectives, controls, and improvement measures. Organizational priorities may differ depending on operational context; for example, institutions processing sensitive personal or research data may emphasize confidentiality, whereas project-oriented systems may prioritize availability and resilience. Since security measures often involve interdependencies and trade-offs among CIA dimensions, such decisions cannot rely solely on ad hoc qualitative assessments [44].

In this context, Multi-Criteria Decision Analysis (MCDA) provides a structured and transparent decision-support approach for ISO/IEC 27001:2022-aligned ISMS implementation. MCDA enables the evaluation of alternative controls and organizational measures against multiple, frequently conflicting criteria related to CIA requirements, as well as economic, technical, and operational considerations. Through systematic weighting and ranking procedures, MCDA supports prioritization of security initiatives, balancing trade-offs and improving resource allocation according to institutional profiles and strategic objectives [45,46]. As a result, MCDA strengthens the practical applicability of ISO/IEC 27001:2022 by integrating qualitative expert knowledge with quantitative decision-support methods, thereby enhancing continual improvement and governance effectiveness in complex information security environments [47].

The hierarchical decision-making structure presented in Figure 2 represents an integrated framework that systematically consolidates the core requirements of ISO/IEC 27001:2022 with supporting standards from the broader ISO/IEC 27000 family.

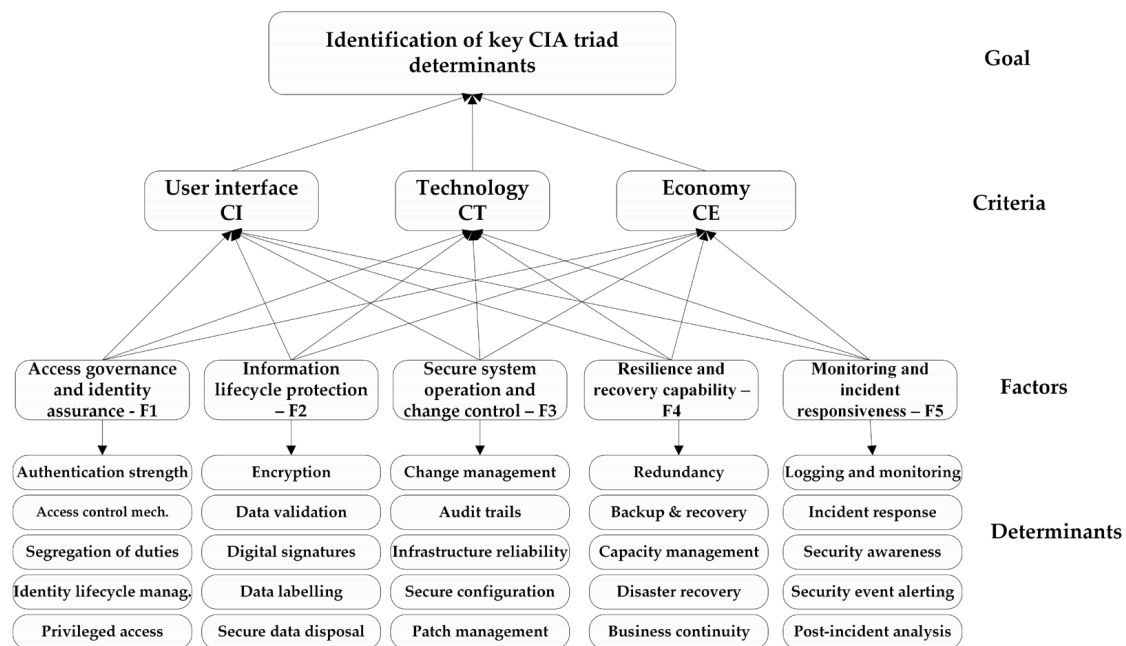


Figure 2. Advisable ISO/IEC 27001:2022 hierarchical decision-making structure. Note: Source: authors.

The user interface criterion (CI) was introduced to capture the human–system interaction dimension of information security, which remains a major source of vulnerabilities despite advances in technical protection measures. ISO/IEC 27001:2022 reinforces the importance of human-centered controls through enhanced awareness, transparency, and error-prevention mechanisms. The technology criterion (CT) represents the technical foundation of information security, including system configuration, cryptographic protection, monitoring, and infrastructure reliability. The revised Annex A places greater emphasis on secure configuration, cloud security, and operational resilience, establishing technology as a key determinant of ISMS effectiveness. The economy criterion (CE) reflects resource availability, cost–benefit considerations, and financial sustainability of security controls. Through threat-based prioritization, ISO/IEC 27001:2022 implicitly recognizes economic feasibility as essential for long-term ISMS implementation and organizational adoption.

F1—Access governance and identity assurance factor synthesizes newly emphasized identity and access management controls in ISO/IEC 27001:2022, reflecting the shift toward lifecycle-based identity governance and privileged access oversight. *F2*—Information lifecycle protection was selected to represent the expanded treatment of information beyond storage and transmission, incorporating classification, handling, validation, and secure disposal. *F3*—Secure system operation and change control addresses the growing complexity of system environments and the need for controlled operational behaviour, configuration stability, and traceable changes. *F4*—Resilience and recovery capability is explicitly strengthened in the ISO 27001:2022 revision through enhanced business continuity and disaster recovery controls. *F5*—Monitoring and incident responsiveness reflects the transition from passive security controls to proactive detection, response, and continuous improvement. The five factor groups (*F1–F5*) were defined to operationalize the confidentiality–integrity–availability (CIA) triad within the MCDA framework. Collectively, these factors preserve the conceptual centrality of the CIA model while enabling its practical evaluation within the proposed MCDA hierarchy.

The determinant groups further operationalize the defined factors. The *F1* determinants (authentication strength, access control mechanisms, segregation of duties, identity lifecycle management, and privileged access) reflect the transition from static authorization models toward dynamic identity governance and accountability mechanisms. The *F2* determinants (encryption, data validation, digital signatures, data labelling, and secure disposal) represent lifecycle-oriented information protection extending beyond traditional perimeter-based controls. *F3* determinants (change management, audit trails, infrastructure reliability, secure configuration, and patch management) address operational complexity and vulnerability management in contemporary information systems. The *F4* determinants (redundancy, backup and recovery, capacity management, disaster recovery, and business continuity) support sustained service delivery and organizational resilience under adverse conditions. Finally, the *F5* determinants (logging and monitoring, incident response, security awareness, security event alerting, and post-incident analysis) capture proactive monitoring, evidence-based incident management, and organizational learning as essential components of continual ISMS improvement.

By integrating core system requirements with complementary guidance, controls, and best practices from the ISO/IEC 27000:2022 series, the proposed hierarchical framework ensures methodological consistency across multiple levels of information security governance [48]. The approach enables integrated consideration of confidentiality, integrity, and availability (CIA) within a unified decision-making structure [49,50], while remaining adaptable to dynamic operational conditions, including peak-load periods such as examination sessions. Furthermore, the defined MCDA hierarchy, particularly the 25 determinants, represents conceptually grounded and expert-informed elements reflecting practical challenges in educational information systems, including scalability demands, identity federation complexity, and device heterogeneity. Consequently, the framework supports realistic prioritization and evaluation of security measures in accordance with internationally recognized ISO practices [51,52].

Priority determinants were identified through a three-level hierarchical structure consisting of three criteria, five factors, and twenty-five determinants presented in Figure 2. Group ranking was performed using the Fuzzy Analytic Hierarchy Process (FAHP), selected due to its ability to address uncertainty and subjective judgment through fuzzy numbers. FAHP is particularly suitable for complex MCDA problems involving uncertainty and the simultaneous evaluation of qualitative and quantitative criteria, which are characteristic of information security governance in educational information systems. Unlike conventional ranking approaches, FAHP enables structured prioritization under conditions where expert judgments are context-dependent and linguistically expressed rather than strictly numerical. The method additionally supports group decision-making and aggregation of multiple expert perspectives, which is fully aligned with the collaborative governance and continual improvement principles embedded within ISO/IEC 27001:2022. To facilitate expert participation and reduce evaluation complexity, variables with predefined fuzzy ranges were incorporated, enabling more intuitive assessment while systematically accounting for uncertainty and subjective interpretation during the decision-making process. Although the resulting prioritization patterns may partially resemble outcomes obtainable through simpler baseline approaches, the proposed FAHP procedure provides important methodological advantages in terms of robustness and consistency of expert-driven evaluation. In particular, the framework maintains methodological stability regardless of the number and heterogeneity of participating experts, making it suitable for complex institutional environments where cybersecurity priorities may differ across organizational roles and operational contexts. Furthermore, in more advanced implementation stages, the framework enables identification of subtler vulnerabilities, interdependencies, and improvement opportunities that are often insufficiently captured through deterministic or

purely quantitative approaches. The hierarchical structuring of determinants together with the linguistic-variable methodology remains fully compatible with classical AHP principles and their fuzzy extensions, allowing assessment of both individual judgment consistency and overall expert agreement. Consequently, the approach contributes to more transparent and evidence-supported development of security management procedures by explicitly incorporating expert consensus into the prioritization and decision-support process. There are numerous scientific applications of AHP and FAHP-like methods in various ISO derived management systems. They have been used in modern management contexts for the selection of sustainable solutions [53], determination of priority actions within OHS systems [54,55], performance evaluation [56,57], support in system transformation [58], information security management systems [59,60] and education information systems [61,62].

In line with standardised FAHP approach, experts define their preferences using triangular fuzzy numbers, and the extent to which the i -th expert influences the decision is determined by the coefficient η_i , which is based on total experience (e_i), experience in information security (c_i), and general experience in IT quality management (q_i):

$$\eta_i = \frac{e_i \cdot c_i \cdot q_i}{\sum_{k=1}^m e_k \cdot c_k \cdot q_k}, \quad (1)$$

for m experts, where i is a natural number less than or equal to m . All variables used in the previous equation quantitatively describe the expert's experience by taking one of the values from the set $\{1,2,3\}$, corresponding to the relevant ranges of years of experience. The lowest value from this set indicates up to four years of experience, while the highest value represents more than eight years of experience.

Experts compare criteria, factors, and determinants in pairs to determine ranks by applying the fuzzy analytic hierarchical process (fuzzy AHP, FAHP). The hierarchical structure of the problem, shown in Figure 2, conditioned the choice of an AHP-like procedure for determining priorities. To incorporate uncertainty into the decision-making process, pairwise comparisons are performed in the fuzzy domain using triangular fuzzy numbers. Fuzzy values ranging from (7,7,9) to (1,1,3) are used to express, respectively, the absolute dominance of one compared alternative over another and the equally important alternatives [63]. Expert k describes his or her preferences when comparing n alternatives using a square matrix

$$\tilde{D}^{(k)} = \left[\tilde{d}_{ij}^{(k)} \right]_{n \times n}, \quad (2)$$

and based on Equation (1), the aggregation of comparisons of m experts is represented by a matrix

$$\tilde{D}_{agg} = \left[\tilde{d}_{ij} \right]_{n \times n}, \quad (3)$$

whose elements are determined as follows [63]:

$$\tilde{d}_{ij \text{ agg}} = \prod_{k=1}^m \left(\tilde{d}_{ij}^{(k)} \right)^{\eta_k}. \quad (4)$$

The aggregate matrix D_{agg} is used to determine the weights in the fuzzy domain

$$\tilde{\omega}_j = \left(\prod_{k=1}^n \tilde{d}_{ik \text{ agg}} \right)^{1/n} / \left(\sum_{i=1}^n \left(\prod_{k=1}^n \tilde{d}_{ik \text{ agg}} \right)^{1/n} \right), \quad (5)$$

for n compared elements, where $\tilde{d}_{ik \text{ agg}}$ denotes the elements of the fuzzy aggregate matrix D_{agg} , with $1 \leq j \leq n$. The consistent matrices of individual experts are aggregated, and

during this process, the consistency of the experts' judgments is evaluated using the centric consistency index (CCI) [64,65]. For square matrices of order four, a CCI value less than 0.35 is considered acceptable, while for square matrices of higher order, the acceptable value is less than 0.37.

Fuzzy criteria weights $\tilde{\omega}_{CI}$, $\tilde{\omega}_{CT}$, and $\tilde{\omega}_{CE}$, are determined based on [52,64] as

$$\tilde{\omega}_{Cj} = \left(\sum_{i=1}^3 \tilde{d}_{ij} \right) / \left(\sum_{i=1}^3 \sum_{j=1}^3 \tilde{d}_{ij} \right). \quad (6)$$

Similarly, factor weights are determined based on their impact on the criteria:

$$\tilde{\omega}_{Fj} = \tilde{F} \otimes \tilde{\omega}_C, \quad (7)$$

where $\tilde{F}$ represents the vector of influence of factors on the criteria, whose elements are determined as

$$\tilde{f}_{ij} = \left(\sum_{j=1}^5 \tilde{d}_{ij} \right) / \left(\sum_{i=1}^5 \sum_{j=1}^5 \tilde{d}_{ij} \right), \quad (8)$$

bearing in mind that five factors are included in the proposed hierarchical structure.

The weights of the determinants are obtained by multiplying the weights of the factors and the value obtained through the comparison of the determinants, that is,

$$\tilde{\omega}_{FkDl} = \left(\left(\sum_{j=1}^5 \tilde{d}_{ij} \right) / \left(\sum_{i=1}^5 \sum_{j=1}^5 \tilde{d}_{ij} \right) \right) \otimes \tilde{\omega}_{Fk}, \quad (9)$$

where $k = 1, \dots, n$ represents the k -th vector, while $l = 1, \dots, n$ represent the determinants.

The fuzzy values of the determinants are then defuzzified by calculating the arithmetic mean of the values, thus obtaining the crisp values of ω_{FkDl} . Further, through normalization, the final crisp weight values ω_{FIDj}^* are obtained:

$$\omega_k^* = \omega_k / (\omega_1 + \omega_2 + \dots + \omega_n), \quad (10)$$

for $k = 1, \dots, n$, where n is the number of compared elements. This ensures that the total sum of the weights equals one. These normalized values are used to compare the impact of individual determinants on the goal. A higher weight value indicates greater importance of the corresponding element in achieving the goal defined at the top of the hierarchical decision-making structure.

Testing of the ranking procedure was performed by introducing minor changes to the weights of the criteria (factors). This analysis examines whether small changes in expert judgments, which are subjective in nature, can lead to significant changes in the final ranking of the determinants. The analysis starts from the obtained rankings and then modifies the weight of the most significant criterion (factor) by a certain percentage. After the modification, the weights of the remaining criteria (factors) are proportionally recalculated to ensure that the sum of weights remains equal to 1. Once the new weights have been determined, the determinants are re-ranked and any changes in their positions are analysed. If the leading determinants retain the same order across greater number of variations (e.g., different percentage of weight change), this confirms the acceptability, or robustness, of the method.

4. Results

4.1. Insights from ISO Survey Data

According to available ISO Survey data (given as Supplementary Material), organizations operating educational information systems are predominantly represented within

the Information Technology sector (48%), reflecting their extensive reliance on learning management systems (LMS), student information systems (SIS), digital repositories, federated identity management, and cloud-based academic services. Educational institutions are also represented within Other Services (11%), as universities, colleges, and private training providers increasingly function as service-oriented organizations, while a smaller proportion falls within Public Administration (3%) due to public-sector governance and regulatory oversight [40]. These classifications highlight the cross-sectoral nature of educational information systems and their dependence on digitally enabled processes. From an information security perspective, the sectoral distribution is particularly relevant because the operational characteristics of EIS closely align with the objectives and requirements of ISO/IEC 27001:2022. The standard emphasizes the establishment of an Information Security Management System capable of protecting information assets through risk-based governance, security controls, and continual improvement processes. Educational information systems routinely process sensitive personal, academic, financial, research, and institutional data, making the preservation of confidentiality essential for privacy protection, integrity critical for ensuring the accuracy and reliability of educational records, and availability necessary for uninterrupted learning and administrative services. The relevance of ISO/IEC 27001:2022 extends beyond higher education. Educational information systems are increasingly integrated into corporate learning and workforce development environments, where by 2025 approximately 83% of large organizations globally utilize EIS platforms for employee training and more than 90% report active LMS adoption to support compliance, professional development, and organizational learning [40]. In such settings, ISO/IEC 27001:2022 provides a structured framework for managing information security risks associated with cloud-based learning platforms, remote access, digital assessments, and knowledge management systems. Consequently, the widespread adoption of EIS across educational and corporate sectors reinforces the importance of CIA-oriented security governance and provides a strong justification for applying the proposed MCDA framework to evaluate and prioritize information security determinants within ISO/IEC 27001:2022-aligned environments.

4.2. MCDA Results

A total of $m = 10$ experts participated in the ranking process. Experts are selected for decision-making at a strategic level, primarily from the fields of education and information security systems (including academic professionals specializing in technical and technological sciences and IT management). Objectivity in the process of ranking determinants was enhanced through the selection of experts with extensive experience from various institutions (University of Niš, Toplica Academy of Applied Studies, Institute for Standardization of Republic of Serbia). All experts met the following criteria: holding a PhD degree, possessing relevant professional experience in the field, having published papers in reputable journals with an Impact Factor, participating in similar research projects, and demonstrating a thorough understanding of the functioning of comparable systems or having active involvement in them. Their extensive expertise was helpful in the process of establishing the hierarchy and developing the procedure for ranking the determinants. Their experience is represented by the following sets of values: $E = \{2,2,2,3,3,2,2,2,3,2\}$, $C = \{2,2,2,2,2,2,2,3,3,2\}$, and $Q = \{2,3,3,3,3,2,3,2,2,3\}$. Based on Equation (1), the experts' weights are $H = \{0.06, 0.09, 0.09, 0.14, 0.14, 0.06, 0.09, 0.14, 0.09\}$, which are then used to determine the aggregate criteria matrix, according to Equation (4):

$$\tilde{D} = \begin{bmatrix} (1, 1, 1) & (0.57, 1.09, 2.02) & (0.53, 1.20, 1.78) \\ (0.50, 0.92, 1.77) & (1, 1, 1) & (0.52, 1.11, 1.69) \\ (0.56, 0.84, 1.90) & (0.61, 0.90, 1.93) & (1, 1, 1) \end{bmatrix}. \quad (11)$$

Further, based on Equation (5), the fuzzy weights of the criteria $w_{c1} = (0.15, 0.36, 0.77)$, $w_{c2} = (0.14, 0.33, 0.72)$, and $w_{c3} = (0.15, 0.30, 0.77)$ are obtained. By defuzzifying these values using Equation (6), the set of crisp criterion values $W_c = \{0.345, 0.323, 0.332\}$ is obtained.

At the next level of the hierarchy, factors are compared in pairs with respect to their importance relative to the criteria. Aggregate matrices are obtained based on Equation (4):

$$\widetilde{CI} = \begin{bmatrix} (1, 1, 1) & (0.53, 0.88, 1.89) & (0.54, 1.01, 1.61) & (0.54, 0.91, 1.74) & (0.56, 0.98, 1.78) \\ (0.53, 1.13, 1.87) & (1, 1, 1) & (0.59, 1.29, 1.92) & (0.67, 1.16, 2.21) & (0.52, 1.01, 1.61) \\ (0.62, 0.99, 1.87) & (0.52, 0.78, 1.69) & (1, 1, 1) & (0.48, 1.02, 1.52) & (0.48, 1.02, 1.43) \\ (0.57, 1.10, 1.85) & (0.45, 0.86, 1.49) & (0.66, 0.98, 2.07) & (1, 1, 1) & (0.61, 1.01, 1.79) \\ (0.56, 1.02, 1.77) & (0.62, 0.99, 1.94) & (0.70, 0.99, 2.10) & (0.56, 1.01, 1.63) & (1, 1, 1) \end{bmatrix}, \quad (12)$$

$$\widetilde{CT} = \begin{bmatrix} (1, 1, 1) & (0.70, 1.08, 2.10) & (0.59, 1.27, 1.86) & (0.46, 0.83, 1.62) & (0.50, 1.09, 1.54) \\ (0.48, 0.93, 1.43) & (1, 1, 1) & (0.44, 0.88, 1.31) & (0.60, 0.90, 1.94) & (0.63, 0.99, 2.04) \\ (0.54, 0.79, 1.69) & (0.76, 1.14, 2.29) & (1, 1, 1) & (0.42, 0.85, 1.31) & (0.55, 0.96, 1.84) \\ (0.62, 1.21, 2.15) & (0.52, 1.11, 1.66) & (0.76, 1.18, 2.41) & (1, 1, 1) & (0.41, 0.92, 1.34) \\ (0.64, 0.92, 2.02) & (0.49, 1.01, 1.58) & (0.54, 1.05, 1.83) & (0.74, 1.09, 2.43) & (1, 1, 1) \end{bmatrix}, \quad (13)$$

$$\widetilde{CE} = \begin{bmatrix} (1, 1, 1) & (0.59, 0.97, 1.84) & (0.64, 1.25, 2.06) & (0.54, 0.90, 1.67) & (0.45, 0.87, 1.39) \\ (0.54, 1.04, 1.68) & (1, 1, 1) & (0.56, 0.94, 1.87) & (0.49, 0.82, 1.64) & (0.45, 0.77, 1.48) \\ (0.49, 0.80, 1.57) & (0.54, 1.07, 1.78) & (1, 1, 1) & (0.40, 0.73, 1.31) & (0.42, 0.84, 1.53) \\ (0.60, 1.12, 1.86) & (0.61, 1.21, 2.03) & (0.76, 1.38, 2.52) & (1, 1, 1) & (0.46, 1.06, 1.48) \\ (0.72, 1.15, 2.23) & (0.68, 1.29, 2.25) & (0.65, 1.19, 2.39) & (0.68, 0.95, 2.18) & (1, 1, 1) \end{bmatrix}. \quad (14)$$

Based on Equation (5), the corresponding fuzzy weights shown in Table 1 are determined. By defuzzifying these fuzzy weights and by determining the mean value of the elements of the fuzzy vector, local crisp values are obtained.

Table 1. Weights of factors ($CCI_{CI} = 0.010$; $CCI_{CT} = 0.009$; $CCI_{CE} = 0.008$).

Factors	Fuzzy Weights	Crisp Weights	Rank
F_1	(0.03, 0.20, 1.12)	0.196	3
F_2	(0.03, 0.20, 1.11)	0.195	4
F_3	(0.03, 0.18, 1.06)	0.185	5
F_4	(0.04, 0.21, 1.18)	0.206	2
F_5	(0.04, 0.21, 1.26)	0.218	1

The same procedure is applied to determine aggregate matrices describing the influence of elements at the lowest level of the hierarchy. The resulting aggregate matrices, obtained using Equation (4), are

$$\widetilde{F1D} = \begin{bmatrix} (1, 1, 1) & (0.34, 0.55, 1.18) & (0.33, 0.84, 1.01) & (0.84, 1.03, 2.53) & (0.86, 1.35, 2.58) \\ (0.84, 1.83, 2.92) & (1, 1, 1) & (0.76, 1.49, 2.64) & (0.90, 1.98, 3.35) & (1.10, 3.04, 4.85) \\ (1.00, 1.19, 2.98) & (0.38, 0.67, 1.31) & (1, 1, 1) & (0.60, 1.11, 1.81) & (0.92, 1.85, 3.16) \\ (0.39, 0.97, 1.18) & (0.31, 0.50, 1.11) & (0.55, 0.90, 1.66) & (1, 1, 1) & (0.78, 1.18, 2.40) \\ (0.39, 0.74, 1.16) & (0.21, 0.33, 0.91) & (0.32, 0.54, 1.09) & (0.42, 0.85, 1.29) & (1, 1, 1) \end{bmatrix}, \quad (15)$$

$$\widetilde{F2D} = \begin{bmatrix} (1, 1, 1) & (0.30, 0.45, 1.18) & (0.41, 1.00, 1.22) & (0.84, 1.10, 2.53) & (0.49, 0.84, 1.53) \\ (0.84, 2.21, 3.28) & (1, 1, 1) & (0.76, 1.89, 2.64) & (1.02, 1.90, 3.48) & (0.85, 1.38, 2.50) \\ (0.82, 1.00, 2.45) & (0.38, 0.53, 1.31) & (1, 1, 1) & (0.70, 1.10, 2.11) & (0.51, 0.86, 1.57) \\ (0.39, 0.91, 1.18) & (0.29, 0.53, 1.01) & (0.48, 0.91, 1.43) & (1, 1, 1) & (0.52, 0.77, 1.53) \\ (0.66, 1.19, 2.06) & (0.40, 0.73, 1.17) & (0.64, 1.17, 1.97) & (0.66, 1.29, 1.97) & (1, 1, 1) \end{bmatrix}, \quad (16)$$

$$\widetilde{F3D} = \begin{bmatrix} (1, 1, 1) & (0.76, 1.04, 2.36) & (0.46, 0.85, 1.38) & (0.42, 0.74, 1.22) & (0.41, 0.80, 1.29) \\ (0.42, 0.97, 1.31) & (1, 1, 1) & (0.54, 0.93, 1.61) & (0.38, 0.72, 1.43) & (0.51, 0.94, 1.53) \\ (0.73, 1.17, 2.18) & (0.62, 1.08, 1.87) & (1, 1, 1) & (0.54, 0.90, 1.78) & (0.54, 1.02, 1.61) \\ (0.82, 1.36, 2.37) & (0.70, 1.39, 2.64) & (0.56, 1.11, 1.86) & (1, 1, 1) & (0.54, 0.88, 1.68) \\ (0.78, 1.25, 2.52) & (0.66, 1.07, 1.97) & (0.62, 0.99, 1.87) & (0.59, 1.13, 1.84) & (1, 1, 1) \end{bmatrix}, \quad (17)$$

$$\widetilde{F4D} = \begin{bmatrix} (1, 1, 1) & (0.27, 0.41, 1.11) & (0.44, 1.02, 1.31) & (0.64, 0.90, 1.93) & (0.49, 0.91, 1.60) \\ (0.90, 2.42, 3.74) & (1, 1, 1) & (0.69, 1.89, 2.63) & (0.84, 1.49, 2.91) & (0.80, 1.44, 2.34) \\ (0.76, 1.01, 2.29) & (0.38, 0.53, 1.45) & (1, 1, 1) & (0.45, 0.74, 1.43) & (0.48, 0.82, 1.43) \\ (0.52, 1.11, 1.55) & (0.34, 0.67, 1.18) & (0.70, 1.36, 2.21) & (1, 1, 1) & (0.60, 1.02, 1.77) \\ (0.63, 1.10, 2.06) & (0.43, 0.71, 1.25) & (0.69, 1.22, 2.10) & (0.57, 0.99, 1.66) & (1, 1, 1) \end{bmatrix}, \quad (18)$$

$$\widetilde{F5D} = \begin{bmatrix} (1, 1, 1) & (0.48, 0.88, 1.43) & (0.51, 0.86, 1.68) & (0.48, 0.92, 1.53) & (0.51, 1.33, 1.69) \\ (0.71, 1.14, 2.10) & (1, 1, 1) & (0.58, 0.99, 1.67) & (0.63, 0.90, 1.82) & (0.75, 1.20, 2.17) \\ (0.59, 1.16, 1.97) & (0.59, 1.01, 1.72) & (1, 1, 1) & (0.58, 1.07, 1.75) & (0.64, 1.46, 2.07) \\ (0.65, 1.09, 2.10) & (0.55, 1.11, 1.60) & (0.57, 0.94, 1.72) & (1, 1, 1) & (0.56, 1.17, 1.90) \\ (0.59, 0.75, 1.98) & (0.46, 0.83, 1.33) & (0.48, 0.69, 1.55) & (0.53, 0.85, 1.78) & (1, 1, 1) \end{bmatrix}. \quad (19)$$

To determine the crisp weights shown in Table 2, the values obtained using Equations (8) and (9) must be normalized. After defuzzification, the weights presented in the table are obtained using Equation (10), and their sum equals one.

Table 2. Weights of determinants ($CCI_{F1C} = 0.010$; $CCI_{F2C} = 0.008$; $CCI_{F3C} = 0.009$; $CCI_{F4C} = 0.008$; $CCI_{F5C} = 0.008$).

Determinants		Fuzzy Weights	Crisp Weights	Rank
F_1D_1	Authentication strength	(0.002,0.034,0.522)	0.035	18
F_1D_2	Access control mechanisms	(0.004,0.066,0.918)	0.061	1
F_1D_3	Segregation of duties	(0.003,0.041,0.651)	0.043	7
F_1D_4	Identity lifecycle management	(0.002,0.033,0.486)	0.032	22
F_1D_5	Privileged access	(0.002,0.024,0.378)	0.025	25
F_2D_1	Encryption	(0.002,0.032,0.490)	0.034	20
F_2D_2	Data validation	(0.004,0.062,0.824)	0.057	3
F_2D_3	Digital signatures	(0.003,0.033,0.557)	0.038	14
F_2D_4	Data labelling	(0.002,0.031,0.419)	0.029	24
F_2D_5	Secure data disposal	(0.003,0.040,0.543)	0.037	16
F_3D_1	Change management	(0.002,0.032,0.462)	0.032	21
F_3D_2	Audit trails	(0.002,0.033,0.451)	0.032	23
F_3D_3	Infrastructure reliability	(0.003,0.038,0.543)	0.038	13
F_3D_4	Secure configuration	(0.003,0.042,0.603)	0.042	8
F_3D_5	Patch management	(0.003,0.040,0.587)	0.041	9
F_4D_1	Redundancy	(0.002,0.033,0.502)	0.034	19
F_4D_2	Backup & recovery	(0.004,0.062,0.832)	0.059	2
F_4D_3	Capacity management	(0.003,0.033,0.545)	0.037	17
F_4D_4	Disaster recovery	(0.003,0.041,0.551)	0.038	15
F_4D_5	Business continuity	(0.003,0.041,0.577)	0.039	12
F_5D_1	Logging and monitoring	(0.003,0.040,0.569)	0.040	11
F_5D_2	Incident response	(0.003,0.043,0.669)	0.047	4
F_5D_3	Security awareness	(0.003,0.046,0.652)	0.046	5
F_5D_4	Security event alerting	(0.003,0.043,0.638)	0.045	6
F_5D_5	Post-incident analysis	(0.003,0.034,0.586)	0.041	10

Based on the results obtained, the local weights of the criteria showed relatively small differences, indicating that experts considered their influence on information security

management to be generally balanced. A similar pattern was observed at the factor level, where no individual factor demonstrated clear dominance. However, at the determinant level, three security measures emerged as the most influential: access control mechanisms ($w_{F1D2} = 0.061$), backup and recovery ($w_{F4D2} = 0.059$), and data validation ($w_{F2D2} = 0.057$). These determinants were therefore analysed in detail within Section 4.3, bearing in mind the proposed improved CIA triad management model. These three determinants represent the highest-ranked operational components within the confidentiality–integrity–availability-oriented decision hierarchy and therefore provide the analytical basis for the proposed CIA triad management model.

The identified determinants can be directly associated with next-generation security protocols relevant to educational information systems. Multi-factor authentication (MFA) aligns primarily with access control mechanisms, reflecting its strong contribution to confidentiality and identity assurance. Lightweight cryptographic mechanisms correspond to data validation, supporting integrity through secure hashing and tamper-evident verification during information exchange processes. Similarly, cryptographic redundancy and secure verification procedures reinforce backup and recovery functions, thereby strengthening availability and protection against unauthorized restoration or data corruption. The obtained weighting results confirm that authentication robustness, cryptographic validation, and recovery capability represent quantitatively prioritized security components rather than isolated technical controls.

In the sensitivity analysis of the applied method, changes were systematically introduced first to the weights of the most significant criterion while keeping the factor weights unchanged, and then to the weight of the most significant factor while keeping the criterion weights unchanged. The weights were varied by the following percentages: $\pm 1\%$ (*Sens1*, *Sens1*–), $\pm 2\%$ (*Sens2*, *Sens2*–), $\pm 5\%$ (*Sens5*, *Sens5*–), and $\pm 10\%$ (*Sens10*, *Sens10*–).

Changes in the weight value of the most significant criterion C_1 produced the results presented in the following table (Table 3).

Table 3. Re-calculated crisp weights of determinants after changing the weight value of the most significant criterion C_1 .

Determinants	<i>Sens10</i> –	<i>Sens5</i> –	<i>Sens2</i> –	<i>Sens1</i> –	<i>Sens1</i>	<i>Sens2</i>	<i>Sens5</i>	<i>Sens10</i>
F_1D_1	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035
F_1D_2	0.061	0.062	0.061	0.061	0.061	0.061	0.061	0.061
F_1D_3	0.043	0.044	0.043	0.043	0.043	0.043	0.043	0.043
F_1D_4	0.032	0.033	0.032	0.032	0.032	0.032	0.032	0.032
F_1D_5	0.025	0.025	0.025	0.025	0.025	0.025	0.025	0.025
F_2D_1	0.033	0.034	0.034	0.034	0.034	0.034	0.034	0.034
F_2D_2	0.057	0.058	0.057	0.057	0.057	0.057	0.057	0.057
F_2D_3	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038
F_2D_4	0.029	0.029	0.029	0.029	0.029	0.029	0.029	0.029
F_2D_5	0.037	0.038	0.038	0.038	0.038	0.038	0.038	0.038
F_3D_1	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032
F_3D_2	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032
F_3D_3	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038
F_3D_4	0.042	0.042	0.042	0.042	0.042	0.042	0.042	0.042
F_3D_5	0.041	0.042	0.041	0.041	0.041	0.041	0.041	0.041
F_4D_1	0.034	0.034	0.034	0.034	0.034	0.034	0.034	0.034
F_4D_2	0.059	0.059	0.059	0.059	0.059	0.059	0.059	0.059
F_4D_3	0.037	0.037	0.037	0.037	0.037	0.037	0.037	0.037
F_4D_4	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038
F_4D_5	0.039	0.039	0.039	0.039	0.039	0.039	0.039	0.039

Table 3. Cont.

Determinants	<i>Sens10</i> –	<i>Sens5</i> –	<i>Sens2</i> –	<i>Sens1</i> –	<i>Sens1</i>	<i>Sens2</i>	<i>Sens5</i>	<i>Sens10</i>
F_5D_1	0.040	0.040	0.040	0.040	0.040	0.040	0.040	0.040
F_5D_2	0.047	0.047	0.047	0.047	0.047	0.047	0.047	0.047
F_5D_3	0.046	0.046	0.046	0.046	0.046	0.046	0.046	0.046
F_5D_4	0.045	0.045	0.045	0.045	0.045	0.045	0.045	0.045
F_5D_5	0.041	0.041	0.041	0.040	0.041	0.041	0.041	0.041

The values of the determined weights did not undergo significant changes, nor did their ranking, which is expected given the defined hierarchical structure and the influence of the criteria weights on the factor weights.

Changes in the weight value of the most significant factor F_5 produced the following results shown in the following table (Table 4).

Table 4. Re-calculated crisp weights of determinants after changing the weight value of the most significant factor F_5 .

Determinants	<i>Sens10</i> –	<i>Sens5</i> –	<i>Sens2</i> –	<i>Sens1</i> –	<i>Sens1</i>	<i>Sens2</i>	<i>Sens5</i>	<i>Sens10</i>
F_1D_1	0.036	0.035	0.035	0.035	0.035	0.035	0.034	0.034
F_1D_2	0.063	0.062	0.061	0.061	0.061	0.061	0.060	0.059
F_1D_3	0.044	0.044	0.043	0.043	0.043	0.043	0.042	0.042
F_1D_4	0.033	0.033	0.032	0.032	0.032	0.032	0.032	0.031
F_1D_5	0.026	0.025	0.025	0.025	0.025	0.025	0.025	0.024
F_2D_1	0.035	0.034	0.034	0.034	0.033	0.033	0.033	0.033
F_2D_2	0.059	0.058	0.057	0.057	0.057	0.057	0.056	0.055
F_2D_3	0.039	0.038	0.038	0.038	0.038	0.038	0.037	0.037
F_2D_4	0.030	0.029	0.029	0.029	0.029	0.029	0.029	0.028
F_2D_5	0.039	0.038	0.038	0.038	0.037	0.037	0.037	0.036
F_3D_1	0.033	0.033	0.032	0.032	0.032	0.032	0.032	0.031
F_3D_2	0.032	0.032	0.032	0.032	0.032	0.031	0.031	0.031
F_3D_3	0.039	0.039	0.038	0.038	0.038	0.038	0.037	0.037
F_3D_4	0.043	0.043	0.042	0.042	0.042	0.042	0.042	0.041
F_3D_5	0.042	0.042	0.041	0.041	0.041	0.041	0.040	0.040
F_4D_1	0.035	0.034	0.034	0.034	0.034	0.034	0.034	0.033
F_4D_2	0.060	0.060	0.059	0.059	0.059	0.058	0.058	0.057
F_4D_3	0.038	0.037	0.037	0.037	0.037	0.036	0.036	0.036
F_4D_4	0.039	0.038	0.038	0.038	0.038	0.037	0.037	0.037
F_4D_5	0.040	0.040	0.039	0.039	0.039	0.039	0.039	0.038
F_5D_1	0.036	0.038	0.039	0.040	0.040	0.041	0.042	0.044
F_5D_2	0.042	0.044	0.046	0.046	0.047	0.048	0.049	0.051
F_5D_3	0.041	0.044	0.045	0.045	0.046	0.047	0.048	0.050
F_5D_4	0.040	0.042	0.044	0.044	0.045	0.046	0.047	0.049
F_5D_5	0.037	0.039	0.040	0.040	0.041	0.041	0.043	0.045

The determinant weight values exhibited minor changes, as did their ranking, in response to the change in changes in the weight of the dominant factor F_5 . This is also evident in the *Sens10*– scenario, where a greater decrease in the weight value of this factor causes it to lose its dominant position. Even in that scenario, the dominant determinants remain in the leading position without any change in ranking and are therefore discussed in greater detail further in the text.

In both cases, the ranking order of the key determinants remained unchanged, which confirms the acceptability (robustness) of the applied method and ranking procedure.

4.3. Improved CIA Triad Management Model—Proposal

4.3.1. Criteria-Based Selection of Determinant-Specific Technologies

While the FAHP analysis prioritized the core security determinants, the selection of specific technological solutions was performed by the authors as an implementation-oriented interpretation of the obtained ranking results. The expert panel was therefore used to evaluate and prioritize security determinants, whereas the subsequent selection of representative technologies was guided by predefined author-defined criteria, relevant literature, ISO/IEC 27001:2022 control requirements, and the operational characteristics of Educational Information Systems (EIS). This distinction is important because the proposed technologies should not be interpreted as an additional expert-ranked output, but as a structured translation of the highest-ranked determinants into feasible implementation mechanisms. Specifically, the three determinants selected for technology specification achieved the following aggregate FAHP weights: access control mechanisms ($w = 0.061$), backup and recovery ($w = 0.059$), and data validation ($w = 0.057$), collectively representing the highest-weighted cluster within the evaluated decision hierarchy.

Given the wide range of available cybersecurity mechanisms, the selection of technologies for each prioritized determinant was guided by five predefined criteria tailored to the operational realities of EIS:

1. ISO/IEC 27001:2022 control alignment.

Selected technologies had to directly operationalize relevant ISO/IEC 27001:2022 Annex A control domains. For example, identity federation, RBAC/ABAC models, MFA, and passwordless authentication support access control, identity management, authentication information, and access rights management. Similarly, parameterized queries, server-side validation, and secure coding practices support application security and secure development, while immutable and offline backup mechanisms support information backup and ICT readiness for business continuity.

2. Direct contribution to CIA dimensions.

Each selected technology had to provide a clear and dominant contribution to at least one CIA dimension. Access control technologies primarily strengthen confidentiality by reducing unauthorized access and privilege misuse. Data validation mechanisms primarily strengthen integrity by preventing malformed, unauthorized, or malicious data from entering academic and administrative workflows. Backup and recovery technologies primarily strengthen availability by supporting restoration of educational services and records after cyber incidents, infrastructure failures, or ransomware attacks.

3. Preference for interoperable and open-standard-based solutions.

Educational institutions frequently operate fragmented and heterogeneous infrastructures composed of legacy student information systems, learning management systems, institutional repositories, cloud services, and externally hosted platforms. In addition, many institutions operate under constrained budgets and cannot rely exclusively on expensive proprietary enterprise platforms. Therefore, preference was given to technologies based on widely adopted open standards, open protocols, and broadly supported implementation patterns, such as SAML, OAuth 2.0, OpenID Connect, FIDO2, JSON-based validation mechanisms, SQL prepared statements, HTTP security headers, and standardized backup principles. This criterion supports cross-platform integration, phased implementation, reuse of existing institutional identity infrastructures, and reduced dependence on restrictive vendor-specific solutions.

4. Relevance to dominant EIS threat scenarios.

Selected mechanisms had to address threat scenarios that are particularly relevant in educational environments, including credential theft, phishing, credential stuffing, excessive privileges, injection attacks, XSS, unauthorized modification of academic records, ransomware, and loss of service availability. For this reason, MFA, FIDO2/passkeys, RBAC/ABAC, and Zero Trust principles were selected for access governance; parameterized queries, schema validation, contextual output encoding, CSP, and WAF mechanisms were selected for application-layer integrity protection; and immutable, air-gapped, geo-redundant, snapshot-based, and orchestrated recovery mechanisms were selected for backup and recovery.

5. Scalability and feasibility in resource-constrained educational institutions.

Academic environments are characterized by high user turnover, annual enrolment cycles, temporary user accounts, heterogeneous user groups, and extensive Bring Your Own Device (BYOD) usage. Therefore, selected technologies had to support phased and scalable implementation without requiring complete redesign of the existing infrastructure. For example, MFA and RBAC can be introduced as baseline controls, identity federation and ABAC can be added to support cross-platform and context-aware access, while immutable backups and automated recovery orchestration can be implemented progressively according to institutional maturity and available resources.

By applying these criteria, the broad FAHP-prioritized determinants were translated by the authors into a focused, implementable, and resource-aware technological stack presented in Tables 5–7. The selected mechanisms should therefore be interpreted as representative determinant-specific implementation options rather than as an exhaustive catalogue of cybersecurity technologies or as an independently ranked set of alternatives. This selection logic strengthens the methodological connection between expert-based prioritization, ISO/IEC 27001:2022-aligned governance requirements, and practical security design decisions in Educational Information Systems.

4.3.2. EIS Threat Context and Core CIA Architecture

Educational institutions are particularly vulnerable to cybersecurity threats due to limited IT resources, budget constraints, and insufficient cybersecurity expertise [66]. Their educational information systems (EIS), including learning management systems, student information systems, repositories, and online assessment platforms, form highly interconnected ecosystems in which cloud adoption and extensive network integration increase the attack surface and complicate security management. Weak authentication and authorization mechanisms are frequently identified as major causes of security breaches as traditional access control approaches often fail to support secure resource sharing in distributed educational environments [67]. Increasing data volumes additionally intensify challenges related to protection of sensitive information and prevention of unauthorized content distribution. Evidence from the UK higher education sector demonstrates recurring access-control-related incidents, including a 2020 password compromise affecting approximately 1000 user accounts and requiring nearly 80 days for recovery [68]. These challenges are further exacerbated by fragmented authentication systems, legacy infrastructure, and user resistance to security measures [69]. Such vulnerabilities are consistent with the OWASP Top Ten Web Application Security Risks (2025), which identifies Broken Access Control as the most critical vulnerability category [70–74].

The security challenges described above are further amplified by the lifecycle characteristics of educational data and the privacy obligations governing its processing. Academic records, assessment results, enrolment data, identity credentials, learning analytics, and activity logs pass through multiple lifecycle stages, including collection, active use, sharing

between institutional systems, archival retention, and scheduled deletion. Each stage introduces distinct confidentiality, integrity, and availability risks, ranging from unauthorized access and excessive retention to inaccurate modification, loss of auditability, or unavailability of critical records during examination and enrolment periods. From a privacy-governance perspective, principles such as data minimization, purpose limitation, access accountability, retention control, and privacy-by-design complement the CIA-oriented objectives of the proposed model. Regulatory frameworks such as the GDPR, and domain-specific educational privacy regulations such as FERPA in the United States, further emphasize the need for controlled access, traceable processing, breach notification, and lifecycle-aware protection of educational data. Therefore, the three highest-ranked determinants—access control, data validation, and backup and recovery—address not only technical threat scenarios, but also the lifecycle-specific risk profile and privacy-governance obligations inherent to educational information systems.

Educational information systems also rely on continuous data exchange among internal platforms and external services, making data validation essential for maintaining system integrity. Inadequate validation mechanisms enable threats such as SQL injection and Cross-Site Scripting (XSS), which are frequently reported in e-learning platforms including Moodle, Chamilo, and ILIAS [70–72]. Security testing using tools such as Acunetix, OpenVAS, and OWASP ZAP has confirmed the presence of XSS and SQL injection vulnerabilities, with injection attacks consistently ranked among the most severe application security threats [73,74]. In parallel, service availability remains critical for continuity of teaching and research activities. Cyber-attacks, hardware failures, and human errors may cause significant disruptions in the absence of effective backup and recovery mechanisms. Increasing ransomware incidents further emphasize the importance of resilient recovery capabilities, with the National Cyber Security Centre identifying educational institutions as frequent ransomware targets [74,75]. Studies additionally indicate that weak backup practices are strongly associated with greater ransomware impacts, while online backups themselves are often compromised during attacks [71,72].

The proposed CIA management model (Figure 3) addresses these challenges through interoperable and standardized security mechanisms, including Security Assertion Markup Language (SAML), OAuth 2.0, OpenID Connect, parameterized queries, and Content Security Policy (CSP) headers. Confidentiality and integrity are strengthened through identity federation, authorization protocols, and application-level validation mechanisms, while availability is supported through Zero Trust principles, immutable backups, and automated recovery procedures. Collectively, these measures reduce exposure to ransomware, injection attacks, and unauthorized access, while supporting operational continuity and resilience. By integrating interoperable technologies with resilience-oriented governance principles, the proposed model provides a scalable and practical information security framework tailored to higher education environments.

4.3.3. Security Operationalization, Interoperability, and Feasibility

The scientific grounding of the selected security determinants, namely access control, data validation, and backup and recovery, is well established in the literature. Evidence from systematic reviews and empirical studies reports consistent improvements across confidentiality, integrity, and availability (CIA) in educational and cloud-based systems [70–77]. For confidentiality, prior studies indicate that multi-factor authentication (MFA) can reduce unauthorized access by over 90%, depending on implementation maturity [78–81]. In specific implementation contexts, Zero Trust Architecture (ZTA) has been reported to reduce insider-threat exposure by around 65% while limiting the attack surface [82,83]. Phishing-resistant methods, such as FIDO2 and biometrics, have also been associated with

reductions in credential theft of up to 85% [84]. Attribute-based access control (ABAC) enhances confidentiality and integrity through fine-grained authorization, with reported improvements of up to 89% in incident response time compared to traditional models [85]. Regarding integrity, data validation is a key defence against application-level attacks. Parameterized queries and prepared statements effectively prevent SQL injection, with some implementations blocking all tested attacks under experimental or controlled conditions [86]. Content Security Policy (CSP) has been reported to reduce XSS attacks by 60–80%, while well-configured rules may improve detection accuracy up to 97% without increasing false positives [87,88]. Additionally, Web Application Firewalls (WAFs) using OWASP rules have been shown to reduce injection and XSS success rates by 60–97%, which is particularly relevant for LMS and SIS environments [89,90]. Regarding availability, advanced backup and recovery strategies are key to digital resilience in educational information systems. Studies show that immutable backups aligned with Zero Trust principles reduce the risk of data loss and ransomware impact [91,92]. Air-gapped backups reduce the risk of malware propagation to recovery copies, while automation enables faster and more coordinated recovery, minimizing downtime [93,94]. Together, these mechanisms form a critical foundation for availability and operational continuity [95]. These reported values are used to support the evidence base for technology selection and should not be interpreted as performance measurements obtained from the proposed framework.

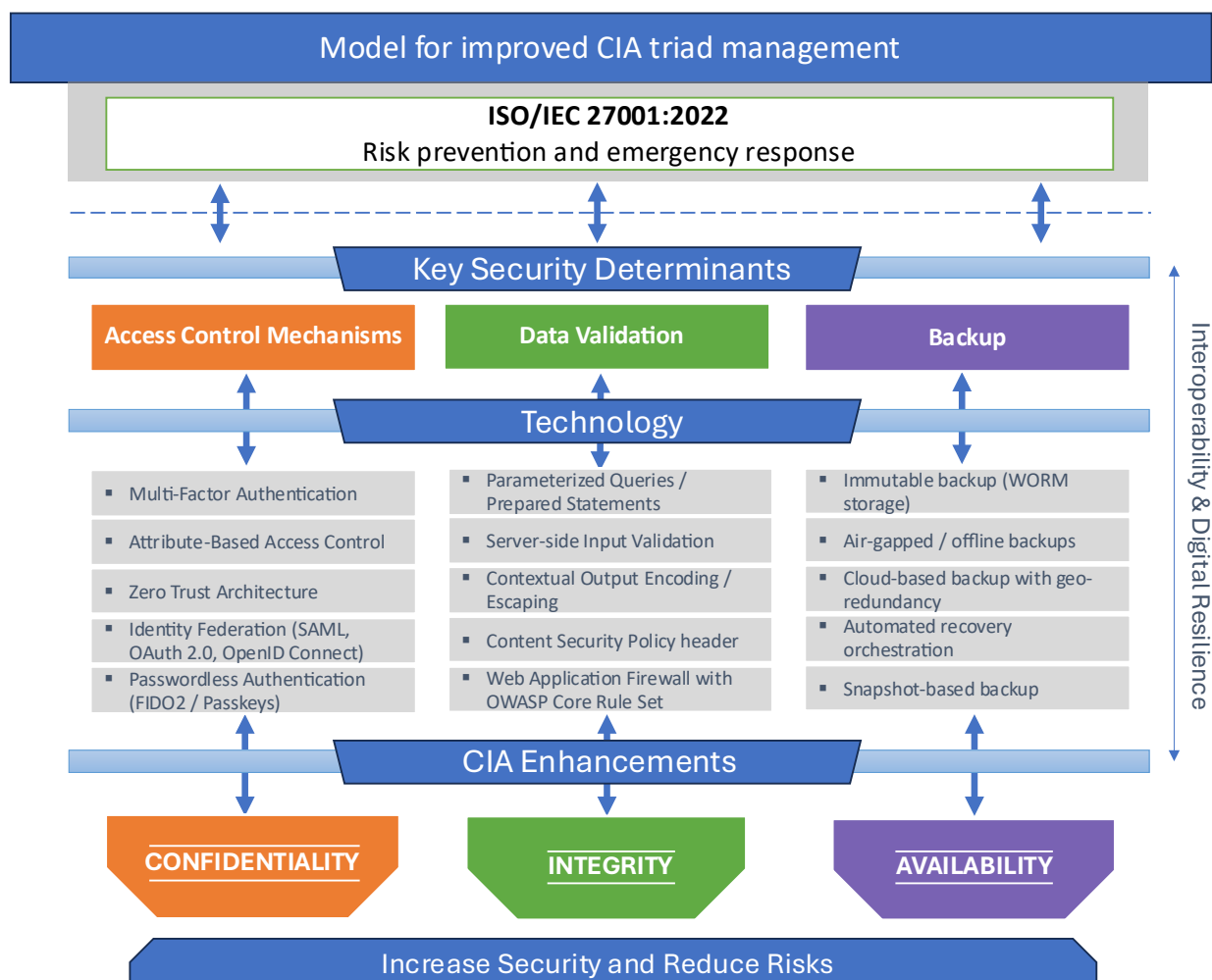


Figure 3. Conceptual security architecture model for reducing cyber risk and enhancing confidentiality, integrity, and availability in educational information systems.

Guided by the previously established selection criteria and supported by this robust scientific grounding, specific technologies and protocols are recommended to operationally enforce the highest-ranked determinants (access control mechanisms $w = 0.061$, backup and recovery $w = 0.059$, and data validation $w = 0.057$). These determinant-specific recommendations, designed to reduce targeted risks while maintaining interoperability, are detailed in Tables 5–7.

Determinant 1 (Access control mechanisms)—recommended technologies:

- Multi-Factor Authentication (MFA);
- Role-Based Access Control (RBAC)/Attribute-Based Access Control (ABAC);
- Zero Trust Architecture (ZTA);
- Identity Federation (SAML, OAuth 2.0, OpenID Connect);
- Passwordless Authentication (FIDO2/Passkeys).

Table 5. Access control mechanisms—recommended technologies for risk reduction.

Technology/Solution	Description	Reduced Risks
Multi-Factor Authentication (MFA)	A supplemental factor in addition to the password (application, SMS, hardware token, biometrics)—significantly increases the security of the account	Phishing, credential stuffing, brute force, account takeover
RBAC + ABAC (Role/Attribute-Based Access Control)	Access by roles and attributes (time, location, study status...)	Unauthorized access, privilege escalation, insider threats, over-privileged accounts
Zero Trust architecture	Every access is verified continuously, without the assumption of trust based on network location	Lateral movement, ransomware breach, insider threats, compromised credentials exploitation
Identity Federation + SSO (SAML, OAuth 2.0, OpenID Connect)	Enables one-time authentication through a central identity provider (institutional account, Google, Microsoft)	Password reuse/weak passwords, phishing on multiple platforms, credential stuffing across multiple profiles
Passwordless Authentication (FIDO2/Passkeys)	Device biometrics or security keys instead of passwords	Phishing, credential theft, brute force, password spraying

Determinant 2 (Data Validation)—recommended technologies:

- Parameterized Queries/Prepared Statements;
- Server-side Input Validation (Whitelist + Schema Checking);
- Contextual Output Encoding/Escaping;
- Content Security Policy (CSP) header;
- WAF (Web Application Firewall) with OWASP Core Rule Set.

Table 6. Data validation—recommended technologies for risk reduction.

Technology/Solution	Description	Reduced Risks
Parameterized Queries/Prepared Statements	Use placeholders instead of string concatenation in SQL/NoSQL queries (e.g., PDO in PHP, PreparedStatement in Java, pymysql in Python 3.13)	SQL injection, NoSQL injection—prevents the execution of malicious code entered through data entry forms, search forms
Server-side Input Validation (Whitelist + Schema Checking)	Strict checking: allowed characters, length, format, data type (regex + JSON Schema/GraphQL schema)	Malformed input → SQL injection, XSS, buffer overflow, logic errors; prevents invalid entries
Contextual Output Encoding/Escaping	Auto-encode before rendering: htmlspecialchars (HTML), json_encode (JS), urlencode (URL), CSS escaping, etc.	Reflected & Stored XSS, DOM-based XSS—protects the display of names, comments, ratings, reports

Table 6. *Cont.*

Technology/Solution	Description	Reduced Risks
Content Security Policy (CSP) header	HTTP header that restricts the sources of scripts, styles, images—prevents the execution of inline/injected JavaScript	XSS (reflected, stored, DOM-based), clickjacking; protects student portals, online exam forms and LMS pages
WAF (Web Application Firewall) with OWASP Core Rule Set	Cloud or on-premise WAF (eg Cloudflare WAF, ModSecurity, AWS WAF). Additional layer—detects and blocks injection/XSS patterns at the application/network level	SQL injection, XSS, command injection, file upload exploits; adds a layer of protection for all web forms in educational systems

For specialized EIS scenarios involving visual or multimedia educational artefacts, such as certificates, scanned examination materials, institutional documents, and repository objects, the integrity mechanisms listed in Table 4 may be complemented by covert information security techniques. Image steganography in color conversion and robust watermarking research provides relevant directions for embedding authenticity markers, supporting tamper detection, and analysing adversarial manipulation of visual educational content [96,97]. These techniques are therefore considered complementary integrity-enhancement mechanisms rather than core determinant-specific technologies within the proposed model.

Determinant 3 (Backup & recovery)—recommended technologies:

- Immutable backup (WORM storage);
- Air-gapped/offline backups;
- Cloud-based backup with geo-redundancy;
- Automated recovery orchestration;
- Snapshot-based backup (cloud-native).

Table 7. Backup & recovery—recommended technologies for risk reduction.

Technology/Solution	Description	Reduced Risks
Immutable backups (WORM/Write Once Read Many snapshots)	Copies that cannot be changed, deleted or encrypted by admins or malware (e.g., SafeMode in Pure Storage, Veeam Immutable, Rubrik)	Ransomware attacks on backups (deletion/modification of copies), insider threats with privileged access, post-compromise data deletion
Air-gapped/offline backups	Physically or logically separated copies (tapes, external disk, enforced isolation)	Ransomware that targets all online copies, total compromise of backup infrastructure
Cloud-based backup with geo-redundancy (Azure Backup, AWS S3 + versioning, Google Cloud Backup)	Automatic off-site with multiple regions and encryption	Local fire/flood/hardware failure, ransomware spreading to local backups, total data centre outage
Automated recovery orchestration (Veeam Orchestrator, Rubrik. . .)	Automatic tests and recovery (failover, runbooks)—reduces human error	Failed recovery, prolonged down-time during exams/registration, undetected backup corruption
Snapshot-based backup (cloud-native + incremental)	Fast point-in-time restore (seconds/minutes) for VMs/databases	Big downtime, data loss from database corruption

The technologies incorporated into the proposed conceptual model were selected through the criteria-based process described above to jointly support interoperability and digital resilience across the confidentiality, integrity, and availability (CIA) dimensions. In one possible implementation pattern, interoperability between the federated identity layer and API authorization layer can be supported through mechanisms such as the SAML Bearer Assertion Grant, where a validated SAML assertion is used to obtain short-

lived, scope-limited API access tokens. This enables institutional SSO to be integrated with API-level authorization without requiring repeated authentication at each EIS service boundary. The practical interaction among these protocol layers within a representative EIS operational scenario is further illustrated through the sequence diagram presented in Section 4.4.

In order to meet the requirement for methodological coherence, the research outcomes are synthesized into a formal mapping framework. To overcome the traditional gap between conceptual security models and operational deployment, a Security Operationalization Matrix (Table 8) was introduced. This matrix represents a formalized translation layer connecting:

- Control requirements from ISO/IEC 27001:2022;
- The top-ranked determinants identified through MCDA analysis;
- Corresponding CIA dimensions (C, I, A weight intensity);
- Recommended technologies and connected EIS-specific operationalization effects.

Table 8. Determinant-Driven Security Operationalization Matrix for ISO/IEC 27001:2022-Aligned CIA Controls.

ISO/IEC 27001:2022 Control Domain	MCDA Determinant (Top Ranked)	CIA Dimension	Recommended Technologies (From Tables 5–7)	EIS-Specific Operationalization Effect
A.5.15 Access control; A.5.16 Identity management; A.5.17 Authentication information; A.5.18 Access rights	F1D2—Access control mechanisms (Rank 1)	Confidentiality (primary), Integrity (secondary)	MFA; Zero Trust Architecture; RBAC/ABAC hybrid model; Identity Federation; Passwordless Authentication.	Enables centralized and context-aware access governance across LMS, SIS, repositories, and cloud services; reduces unauthorized access, credential misuse, and privilege escalation in high-turnover academic environments
A.8.13 Information backup; A.5.30 ICT readiness for business continuity; A.5.29 Information security during disruption	F4D2—Backup & recovery (Rank 2)	Availability (primary), Integrity (secondary)	Immutable backup; Air-gapped backups; Cloud-based backup with geo-redundancy; Automated recovery orchestration; Snapshot-based backup.	Supports continuity of teaching, assessment, enrolment, and administrative services; reduces downtime and improves restoration integrity after ransomware, system failure, or accidental data loss
A.8.28 Secure coding; A.8.29 Security testing in development and acceptance; A.8.20 Network security; A.8.26 Application security requirements	F2D2—Data Validation (Rank 3)	Integrity (primary), Availability (secondary)	Parameterized Queries; Server-side Input Validation; Contextual Output Encoding/Escaping; Content Security Policy (CSP) header; WAF.	Protects academic and administrative transactions from injection, XSS, malformed payloads, and unauthorized data modification; supports integrity of grades, enrolment records, assessment data, and institutional repositories

The individual security mechanisms included in the proposed model, such as MFA, RBAC/ABAC, identity federation, parameterized queries, CSP, WAF, immutable backups, and recovery orchestration, are not presented as novel technologies in isolation. These mechanisms are already well established in cybersecurity and ISMS literature. The contribution of this study lies instead in their structured, determinant-driven integration within

an Educational Information System context. Specifically, the proposed matrix translates the three highest-ranked FAHP determinants into ISO/IEC 27001:2022-aligned control domains, dominant CIA contributions, recommended implementation mechanisms, and EIS-specific operationalization effects.

This translation is intended to address a practical gap frequently observed in educational institutions: the difficulty of moving from generic ISO/IEC 27001:2022 control requirements and abstract CIA objectives toward prioritized, implementable, and context-sensitive security design decisions. In this sense, the matrix does not merely restate known ISMS controls; rather, it functions as an operationalization layer that connects expert-based prioritization, CIA-oriented risk interpretation, ISO control alignment, and technology selection for heterogeneous EIS environments.

The novelty of the matrix is therefore integrative and decision-support-oriented. It provides a structured mechanism for determining which security controls should be implemented first, why they are relevant to a specific CIA dimension, how they correspond to ISO/IEC 27001:2022 control areas, and which open-standard-based or modular technologies can support their implementation. This contribution is particularly relevant for educational institutions, where cybersecurity decisions are constrained by fragmented infrastructures, high user turnover, BYOD usage, limited budgets, and uneven cybersecurity maturity.

The matrix operationalizes governance logic by transforming management-level security objectives into implementable technical controls. Specifically, the access control determinant (*F1D2*) is linked to ISO/IEC 27001:2022 controls related to access control, identity management, authentication information, and access rights. The backup and recovery determinant (*F4D2*) is linked to information backup, information security during disruption, and ICT readiness for business continuity. The data validation determinant (*F2D2*) is linked to secure coding, application security requirements, security testing, and related technical protection controls. This structured mapping eliminates interpretative ambiguity between strategic security planning and technical execution, supporting resource-efficient and priority-driven security governance in EIS environments.

While the Security Operationalization Framework defines the conceptual and control-level mapping between FAHP-prioritized determinants, ISO/IEC 27001:2022 control domains, CIA dimensions, and recommended technologies, an additional implementation-oriented interpretation is required to address the practical constraints of educational institutions. Therefore, Table 9 summarizes feasibility and resource considerations related to the same determinant-specific technology groups, focusing on implementation prerequisites, infrastructure and licensing implications, operational maintenance, and cost-reduction strategies.

The mapping indicates that the proposed framework does not require simultaneous deployment of all mechanisms. Instead, institutions may adopt the technologies progressively, starting from mechanisms that rely on existing application code, identity infrastructure, and backup procedures, and then extending toward more mature controls such as ABAC, FIDO2/passkeys, Zero Trust enforcement, immutable backup, and automated recovery orchestration. This phased implementation logic directly addresses resource constraints while preserving alignment between FAHP-prioritized determinants, CIA objectives, and ISO/IEC 27001:2022-oriented governance.

Table 9. Implementation Feasibility and Resource Considerations for Recommended CIA Determinant-Specific Technologies.

MCDA Determinant	Recommended Technologies	Implementation Prerequisites	Licensing and Infrastructure Model	Operational Maintenance	Feasibility and Cost-Reduction Strategy
<i>F1D2: Access control mechanisms—Confidentiality</i>	MFA, RBAC/ABAC, Identity Federation using SAML/OIDC, FIDO2/Passkeys for high-risk transactions, Zero Trust principles	Integration with an existing Identity Provider; definition of institutional roles, attributes, and authorization policies; identification of high-risk academic transactions	Can rely on open-standard-based or open-source IdP solutions, existing institutional accounts, or academic identity federations; reduces dependence on proprietary identity platforms	Periodic access-policy audits, user lifecycle management for students and staff, temporary account management, MFA reset procedures, and review of privileged access	Reuses existing academic identity infrastructure where available. Primary investment is shifted from proprietary licensing toward engineering integration, policy definition, and identity lifecycle governance
<i>F2D2: Data validation—Integrity</i>	Parameterized queries, server-side schema validation, contextual output encoding, CSP headers, WAF with OWASP Core Rule Set	Refactoring of legacy forms and database access layers; definition of validation schemas and business rules; configuration of HTTP security headers and WAF rules	Can rely on native framework mechanisms, database APIs, JSON schema validation, web server configuration, and open-source WAF engines such as ModSecurity with OWASP CRS	Continuous monitoring of WAF logs, false-positive tuning, CSP report analysis, secure coding review, and updates of validation rules according to application changes	Uses built-in development framework capabilities and community-maintained security rulesets, reducing the need for additional dedicated hardware or vendor-specific application security tools
<i>F4D2: Backup and recovery—Availability</i>	Immutable or WORM-based backup, air-gapped/offline backups, snapshot-based recovery, cloud or off-site geo-redundant backup, automated recovery orchestration	Definition of RPO/RTO targets; backup policy configuration; provisioning of isolated or immutable storage; recovery runbooks and periodic restoration testing	Can combine existing institutional storage, NAS snapshot capabilities, open-source or commercial backup tools, and pay-as-you-go cloud storage where appropriate	Automated recovery testing, capacity monitoring, integrity checks of backup copies, disaster recovery drills, and periodic review of retention policies	Supports phased implementation by combining existing storage resources with selective use of immutable, offline, or cloud-based recovery layers. Tiered storage and snapshot-based approaches help align resource use with actual data volume and criticality

4.3.4. Manageable CIA Implementation Roadmap Proposal

The proposed CIA implementation roadmap (Figure 4) provides a manageable and operationally grounded structure for the progressive deployment of confidentiality, integrity, and availability controls within EIS. Organized across five sequential phases, from foundational ISMS definition to advanced CIA model integration, the roadmap translates standard requirements into actionable steps that reflect the specific conditions of EIS. By explicitly incorporating EIS challenges the roadmap ensures that security implementation is not treated as a static compliance exercise but as a context-aware and adaptive process. Furthermore, the integration of MCDA-based prioritization mechanisms in the final phase reinforces the empirical backbone of the model, enabling institutions to align control implementation with operational constraints and strategic priorities in a transparent and methodologically consistent manner.

The proposed roadmap demonstrates how a phased and CIA-oriented structuring of security controls can support the transition from baseline compliance toward dynamic governance and sustainable ISMS operation. The clear delineation of milestones across confidentiality, integrity, and availability domains facilitates both vertical (within-domain) and horizontal (cross-domain) integration of ISO 27001:2022 security controls, while also

enabling iterative evaluation in line with the Plan-Do-Check-Act cycle. Importantly, the roadmap operationalizes the concept of adaptability by embedding feedback loops and optimization steps that allow institutions to respond to evolving threats and system demands. The inclusion of EIS-specific adaptations and MCDA-driven prioritization further strengthens its practical applicability, addressing the limitations of generic frameworks and supporting resource-efficient decision-making. As such, the roadmap contributes to the advancement of institutional ISMS sustainability by linking standard compliance with continuous improvement and long-term resilience of information security governance. Also, three-layer roadmap structure explicitly operationalizes all three CIA dimensions through determinant-specific mappings: confidentiality through adaptive access governance, integrity through validation and verification mechanisms, and availability through resilient backup and recovery architectures.

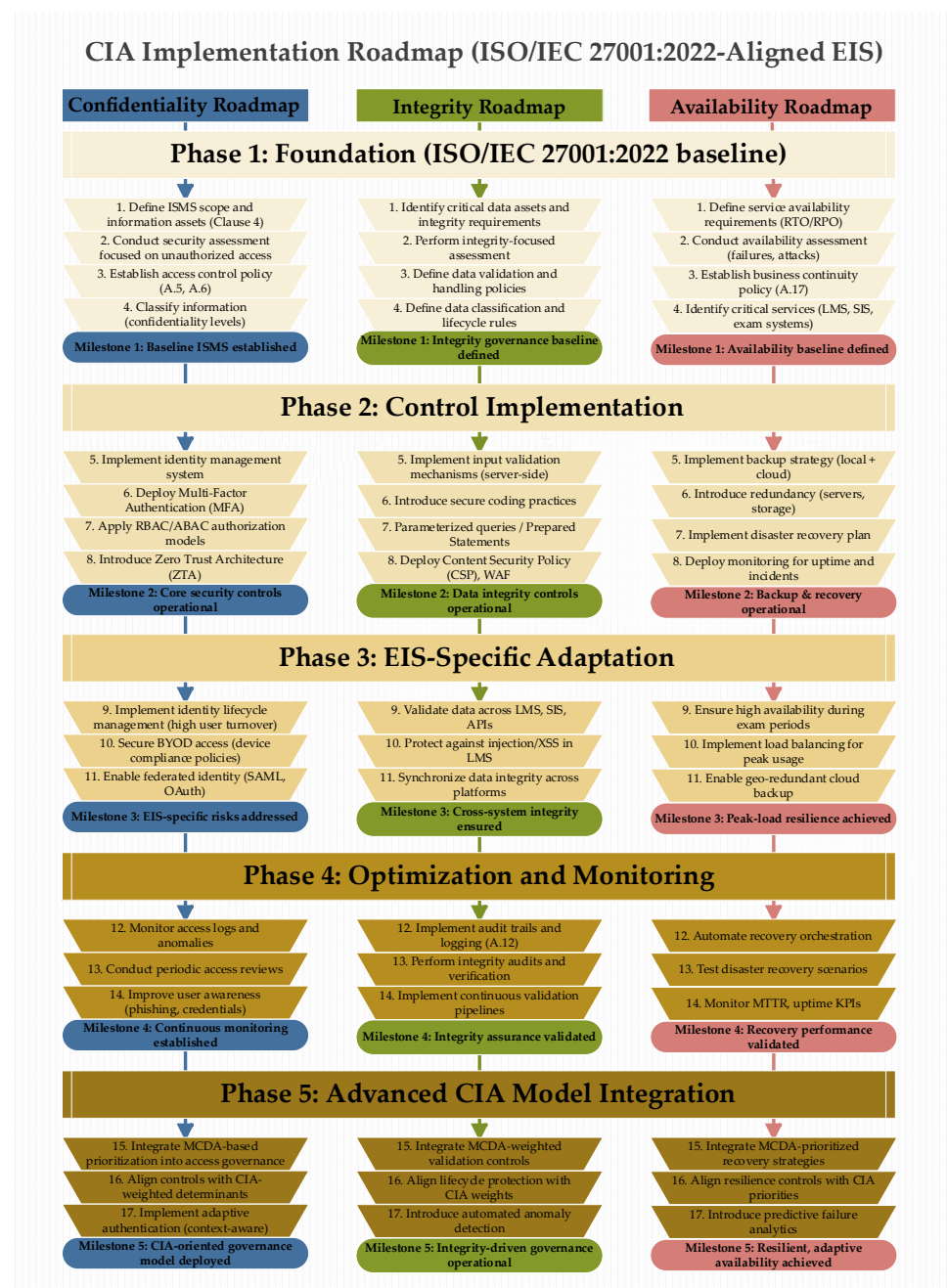


Figure 4. Manageable CIA implementation roadmap.

4.4. Illustrative Technical Operationalization of the Proposed CIA Framework

4.4.1. Conceptual Design-Level Operationalization

The implementation artifacts presented in this section represent conceptual design-level operationalization rather than production-environment code snippets or a fully deployed system. Their primary purpose is to demonstrate the architectural feasibility, internal coherence, and application-layer applicability of the proposed CIA model. Rather than serving as an exhaustive implementation blueprint, Figures 5 and 6 illustrate how the FAHP-prioritized determinants can be translated into architectural layers, workflow steps, and determinant-specific security mechanisms.

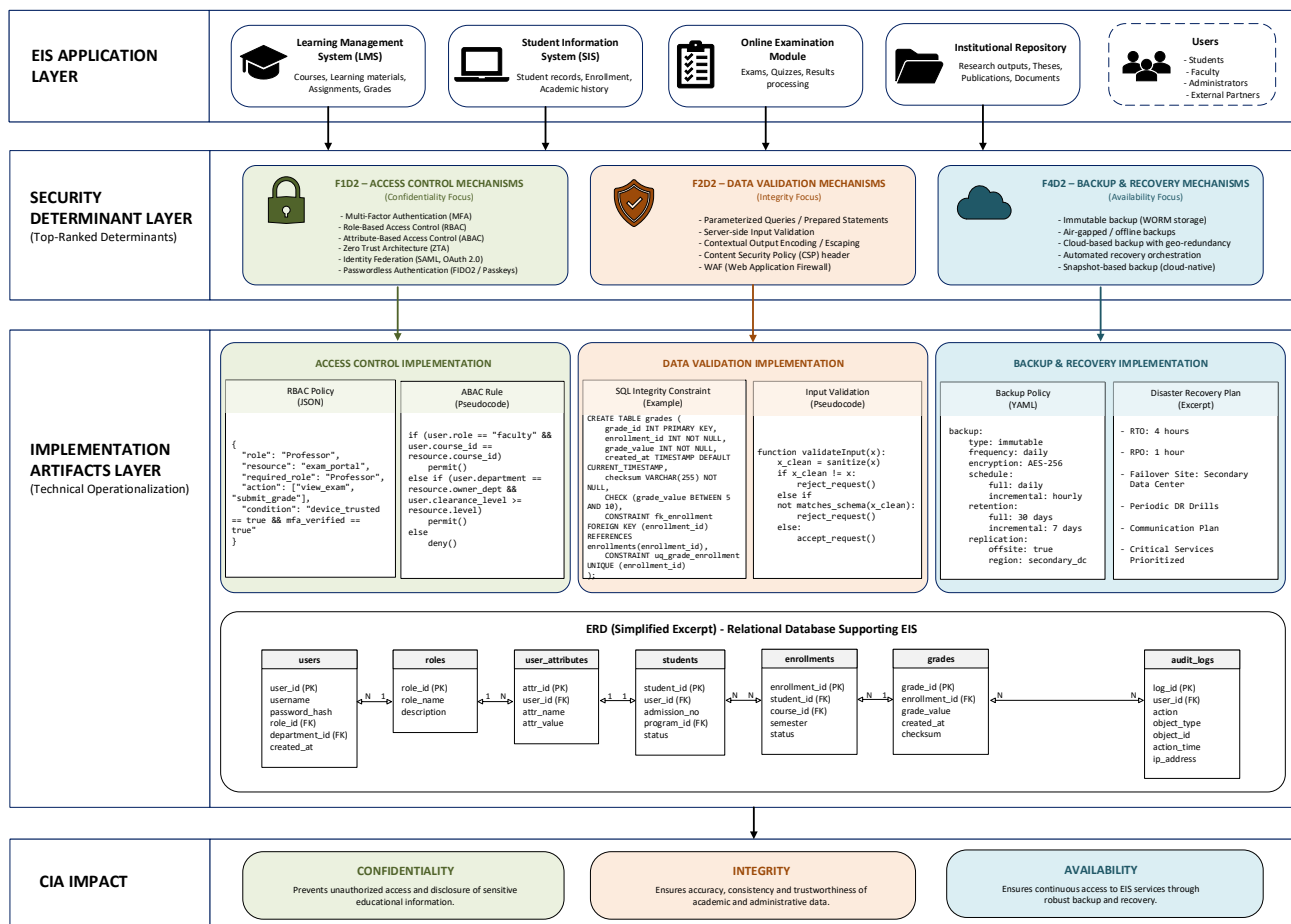


Figure 5. Determinant-Driven CIA Operational Architecture for EIS.

Figure 5 presents the determinant-driven operational architecture for EIS. It connects access control mechanisms (F1D2), data validation (F2D2), and backup and recovery (F4D2) with application-layer enforcement components. In this architecture, access control is not treated only as a login function, but as an identity and authorization enforcement layer where data validation is not limited to user-interface checks but is positioned as an integrity-preserving middleware, and database-level control and backup and recovery are interpreted as availability and resilience mechanisms rather than passive archival functions.

The illustrative fragments included in Figure 5 should be interpreted as determinant-specific design artifacts rather than production-ready code. For the access control determinant (F1D2), the RBAC policy and ABAC rule demonstrate how authorization decisions can be conditioned not only on static roles, but also on contextual attributes such as role type, departmental ownership, clearance level, device trust, and MFA verification status. This design supports adaptive access governance and reduces the likelihood of privilege

misuse, unauthorized access to academic services, and inappropriate access by users outside the relevant institutional context. However, its practical effectiveness depends on the completeness, accuracy, and timely maintenance of identity attributes, role assignments, departmental mappings, and device-trust signals within the institutional identity provider.

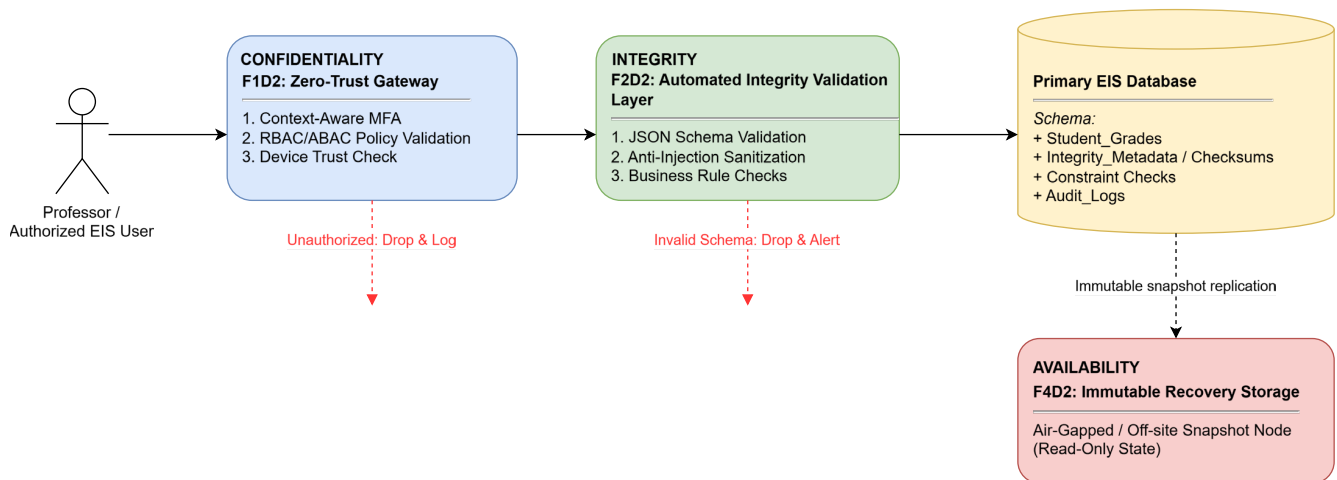


Figure 6. Transaction-Level Workflow for CIA Enforcement During Grade Submission in EIS.

For the data validation determinant (F2D2), the SQL integrity constraint and input validation pseudocode demonstrate a layered integrity-protection approach. The database-level constraint enforces referential integrity and acceptable grade ranges, while the validation middleware sanitizes input and rejects malformed or schema-incompatible requests before they reach the persistence layer. This combination reduces the possibility that compromised user interfaces or vulnerable API endpoints could be exploited for SQL injection, malformed payloads, or unauthorized manipulation of academic records. Nevertheless, the effectiveness of this layer depends on the completeness of validation schemas, correct use of parameterized transactions, and consistent enforcement of business rules across all EIS entry points.

For the backup and recovery determinant (F4D2), the backup policy and disaster recovery excerpt demonstrate availability-oriented resilience through immutable backup configuration, encrypted daily and incremental backup scheduling, off-site replication, defined retention periods, RTO/RPO targets, failover planning, periodic recovery drills, communication procedures, and prioritization of critical services. A critical design characteristic is the logical or administrative isolation of recovery storage and backup credentials from the primary operational environment, which reduces the possibility that ransomware processes or compromised privileged accounts can modify or delete backup copies. The effectiveness of this layer depends on regular restoration testing, monitoring of backup integrity, capacity planning, and alignment between configured recovery objectives and institutional service requirements.

Figure 6 extends this architectural view by presenting an operational workflow for a high-risk academic transaction, namely grade submission to the Student Information System. This workflow is suitable for illustrating CIA enforcement because the transaction simultaneously requires confidentiality, integrity, and availability. Confidentiality is required because only an authorized professor or administrative user should be permitted to access and submit grades. Integrity is required because grade values, course ownership, student enrolment, and exam-period constraints must be validated before committing the transaction. Availability is required because academic records must remain recoverable and operationally accessible after infrastructure failures, accidental data loss, or ransomware incidents.

The workflow therefore demonstrates that the proposed model does not implement the CIA dimensions as isolated control groups. Instead, confidentiality, integrity, and availability

are applied sequentially during the lifecycle of a high-risk academic transaction: access control determines whether the transaction may be initiated, validation determines whether the transaction may be accepted, and recovery mechanisms determine whether the transaction can be restored without compromise. This interpretation strengthens the connection between the FAHP-prioritized determinants and their practical operational role in EIS.

4.4.2. Sequence-Level Interoperability Demonstration

While Figure 6 presents the conceptual workflow at the transaction level, Figure 7 decomposes the same grade-submission scenario into protocol-level and component-level interactions. This sequence-level view further substantiates the interoperability of the proposed security mechanisms within a heterogeneous academic environment by showing how external federated identity management based on SAML can be integrated with internal API authorization mechanisms based on OAuth 2.0-compatible access tokens and JSON Web Tokens (JWT).

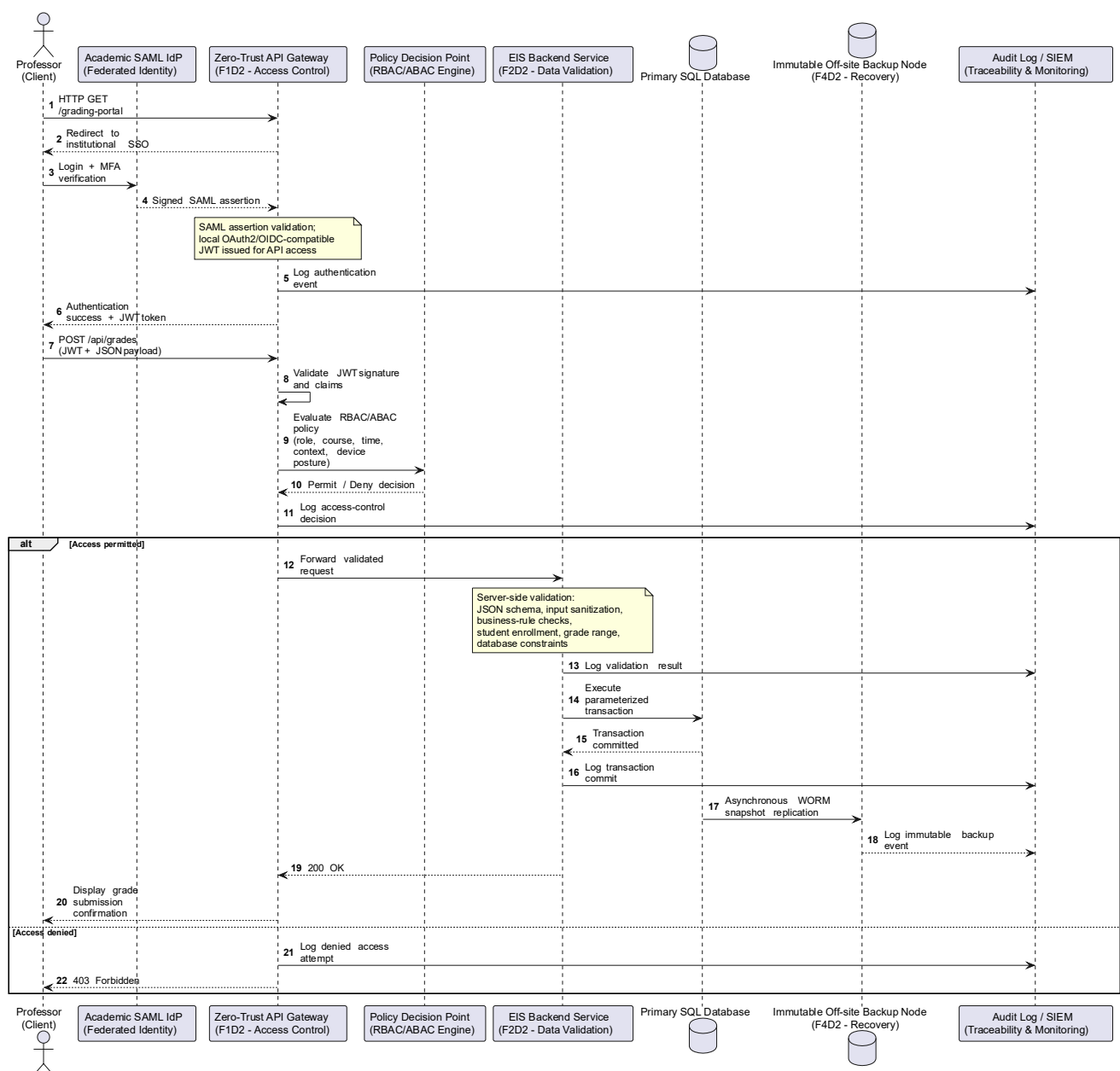


Figure 7. Sequence-Level Interoperability Demonstration for CIA Enforcement in an EIS Grade Submission Process.

The conceptual architecture separates policy decision-making and policy enforcement by introducing a Policy Decision Point (PDP). The Zero-Trust API Gateway validates the external SAML assertion and issues an internal, short-lived, scope-limited API authorization token only after the PDP evaluates RBAC/ABAC rules, such as user role, course ownership, student enrolment, transaction sensitivity, and exam-period timeframes. This design supports contextual access control without requiring repeated authentication at each EIS service boundary.

The workflow also integrates a centralized Security Information and Event Management (SIEM) or audit logging component. By logging critical events, including authentication outcomes, authorization decisions, validation results, database commits, and backup replication events, the model supports traceability and post-incident analysis. This is particularly important in educational environments because high-risk academic transactions, such as grade submission, must be auditable and attributable without exposing unnecessary personal or operational data.

Consequently, the sequence-level model demonstrates how identity assurance, contextual authorization, data validation, transactional database processing, immutable recovery replication, and audit logging can operate as a coherent security pipeline. This directly addresses the interoperability challenge by showing how the access control determinant (*F1D2*), data validation determinant (*F2D2*), and backup and recovery determinant (*F4D2*) interact during a single EIS transaction.

4.4.3. Scope, Limitations, and Future Validation Outlook

It should be explicitly noted that the current phase of this research is focused on conceptual mapping, architectural feasibility, and MCDA-driven prioritization of CIA-oriented controls. The proposed workflows and architectural artifacts serve as a design-level feasibility demonstration that bridges the gap between abstract ISO/IEC 27001:2022 requirements and concrete engineering tasks. They should not be interpreted as evidence of a completed production deployment or as longitudinal empirical validation.

While the framework is logically designed to mitigate dominant vulnerabilities within Educational Information Systems, verifying its practical effectiveness in complex real-world network environments requires future empirical deployment. Such validation should include longitudinal observation, production-scale monitoring, incident analysis, availability measurement, and comparative evaluation before and after framework adoption. A 3- to 6-month pilot study in one or more educational institutions would therefore represent a logical next step.

To avoid overstating the empirical status of the proposed operationalization, Table 10 does not report measured production results. Instead, it defines scenario-based evaluation indicators that can be used in future pilot deployments to assess whether the determinant-specific mechanisms illustrated in Figures 5–7 operate as intended. The indicators are aligned with the same FAHP determinants, CIA dimensions, and architectural components used throughout the proposed model.

This evaluation outlook addresses the limits of the present conceptual study without overstating its empirical claims. It also provides measurable directions for future pilot deployment, including incident reduction rates, recovery time objectives, recovery point objectives, audit completeness, and system availability during high-demand academic periods such as enrolment, examination sessions, and grade submission deadlines.

Table 10. Scenario-Based Evaluation Indicators for Future Validation of the Operationalized CIA Framework.

CIA Dimension	FAHP Determinant	Scenario-Based Evaluation Check	Related Mechanism	Possible Indicator
Confidentiality	<i>F1D2</i> —Access control mechanisms	Attempted grade submission by an unauthorized user or from an unusual context	MFA, RBAC/ABAC, PDP, Zero-Trust Gateway, device-trust evaluation	Blocked unauthorized requests; denied policy decisions; anomalous access attempts
Confidentiality/ Integrity	<i>F1D2</i> —Access control mechanisms	Use of valid credentials outside authorized course ownership or exam period	RBAC/ABAC policy evaluation, course ownership validation, time-based access constraints	Number of denied context-invalid transactions; policy violation rate
Integrity	<i>F2D2</i> —Data validation	Submission of malformed grade payload, out-of-range value, or injection attempt	Schema validation, sanitization, parameterized queries, database constraints	Rejected malformed transactions; blocked injection payloads; validation error rate
Integrity	<i>F2D2</i> —Data validation	Attempted unauthorized modification of previously submitted academic records	Business-rule validation, checksum verification, audit logging, database constraints	Detected unauthorized modification attempts; audit trail completeness
Availability	<i>F4D2</i> —Backup and recovery	Simulated database failure or ransomware-related recovery scenario	Immutable backup, snapshot recovery, off-site replication, DR drills, RTO/RPO planning	Recovery success rate; achieved RTO; achieved RPO; integrity of restored records; backup immutability verification
Governance	Cross-cutting	Reconstruction of transaction history after incident or dispute	SIEM, audit logs, PDP decision logs, event correlation	Completeness of authentication, authorization, validation, commit, and backup logs

5. Discussion

The increasing dependence of educational institutions on digital services, together with the growing sophistication of cyber threats, necessitates more structured and scalable approaches for securing educational information systems. Similar to other sectors, education environments are exposed to phishing, malware, ransomware, unauthorized access, and data integrity breaches. These challenges justify the development of a conceptual framework with implementation-oriented decision-support elements that integrates ISO/IEC 27001:2022 principles, digital threat-management processes, and targeted security mechanisms to strengthen the confidentiality, integrity, and availability (CIA) of educational data and services. Although educational information system security depends on numerous technical and organizational factors, both the literature and expert evaluation indicate that access control mechanisms, data validation, and backup and recovery processes represent the highest-priority determinants for effective CIA-oriented security management. Among these determinants, access control mechanisms emerged as the most influential because they represent the primary safeguard against unauthorized access and frequently condition the effectiveness of other protection measures. Consistent findings in the literature further

identify weaknesses in access governance as one of the dominant causes of educational information system compromise.

From an interpretative perspective, the prioritization of access control mechanisms reflects the broader transition from perimeter-oriented security toward identity-centric and context-aware protection models in modern educational environments. The high weighting assigned to the access control determinant emphasizes the strategic role of layered authentication mechanisms in reducing credential-compromise risks. Similarly, the significance of data validation highlights the increasing importance of integrity-oriented controls within distributed, cloud-supported, and API-driven academic infrastructures. The relevance of cryptographically secured backup and recovery processes additionally demonstrates that availability management is closely interconnected with integrity assurance. Collectively, these relationships indicate that contemporary security protocols are structurally embedded within the prioritized CIA determinants, supporting the internal coherence and practical relevance of the proposed framework for digitally intensive educational ecosystems.

The operationalization presented in this study shows that the contribution of the proposed framework does not lie in introducing new standalone security technologies, but in providing a determinant-driven translation mechanism from FAHP results to ISO/IEC 27001:2022-aligned controls, recommended technologies, feasibility considerations, and scenario-based validation indicators. In this sense, the Security Operationalization Matrix, implementation feasibility mapping, and the sequence-level interoperability model illustrated in Figure 7 jointly bridge the gap between generic ISMS requirements and implementable security design decisions in educational institutions. This is particularly relevant for institutions operating under budget constraints, fragmented identity infrastructures, legacy applications, high user turnover, and uneven cybersecurity maturity. The scenario-based evaluation indicators further provide a structured basis for future pilot validation without presenting the current conceptual framework as a completed production deployment.

Although the framework was developed specifically for Educational Information Systems (EIS), its determinant-based architecture and MCDA-driven prioritization logic may also provide conceptual applicability to other knowledge-intensive sectors characterized by regulated data flows and multi-stakeholder governance. Potential adaptation areas include corporate training systems, quality-assurance learning environments, cybersecurity training platforms, and government-supported educational infrastructures where confidentiality, integrity, and availability of digital learning resources remain strategically important. Nevertheless, the framework is intentionally tailored to higher education environments, particularly those characterized by heterogeneous technical capacities and resource constraints. Since the present study focuses on a conceptual and implementation-support framework rather than a fully deployed operational system, future research should emphasize pilot implementations, comparative validation, and longitudinal assessments conducted before and after framework adoption in real institutional environments.

6. Conclusions

The continuous global increase in ISO/IEC 27001:2022 certifications reflect the growing recognition of information security management as a strategic organizational requirement rather than solely a technical obligation. Within this context, the present study contributes through the systematic identification and prioritization of key determinants influencing the confidentiality, integrity, and availability (CIA) triad in Educational Information Systems (EIS) using an MCDA/FAHP-based framework. By extending the CIA triad from a purely conceptual principle toward a structured implementation-support perspective, the study emphasizes access control mechanisms, data validation processes, and backup and recovery capabilities as

the most influential determinants of information security effectiveness in educational environments. The proposed prioritization framework additionally enables transparent evaluation of technological, organizational, and operational interdependencies that are frequently addressed separately within conventional ISMS assessments. More specifically, the framework translates prioritized CIA determinants into ISO/IEC 27001:2022-aligned control domains, technology recommendations, implementation feasibility considerations, sequence-level interoperability logic, and scenario-based validation indicators.

The findings identify access control mechanisms as the dominant determinant of confidentiality, highlighting the strategic importance of identity governance and authentication processes in preventing unauthorized access to sensitive educational information. Integrity is primarily associated with data validation mechanisms that support the accuracy and trustworthiness of academic and administrative data, while availability depends strongly on resilient backup and recovery strategies capable of maintaining continuity during cyber incidents or system disruptions. Together, these determinants demonstrate that effective educational information security requires coordinated technical and managerial controls aligned with ISO/IEC 27001:2022 principles. In this regard, the proposed framework provides an implementation-oriented structure capable of strengthening security while remaining adaptable to heterogeneous educational infrastructures, legacy systems, fragmented identity environments, and limited institutional resources. Furthermore, the study includes an illustrative operationalization pattern intended to demonstrate how prioritized determinants may be translated into practical security mechanisms within educational information systems. However, this component should be interpreted as an example of applicability rather than as a fully validated operational deployment model.

Several limitations should nevertheless be acknowledged. The proposed MCDA/FAHP framework remains dependent on expert judgment, contextual assumptions, and institutional variability, which may influence determinant weighting and prioritization outcomes. In addition, educational information systems operate within highly dynamic environments characterized by diverse user groups, frequent onboarding and offboarding processes, decentralized data ownership, BYOD practices, and intensive usage fluctuations. The framework addresses these conditions conceptually through adaptive access governance, contextual authentication, application-layer validation, and resilient recovery mechanisms; however, empirical implementation and quantitative performance evaluation remain outside the scope of the present study.

Accordingly, future research should focus on pilot implementations, comparative institutional validation, and longitudinal assessment of ISO/IEC 27001:2022-aligned security improvements within real educational environments. Additional integration with privacy governance frameworks and sustainability-oriented digital governance approaches may further extend the applicability of the proposed framework. Overall, the study contributes to the evolving discourse on information security management by providing a determinant-driven, FAHP-supported, and ISO/IEC 27001:2022-aligned decision-support framework that links CIA prioritization, security technology selection, implementation feasibility, and future validation planning for educational information systems.

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Abbreviations

The following abbreviations are used in this manuscript:

ISO	International Standard Organization
ISS	Institute for Standardization of Serbia
ISMS	Information Security Management System
CIA	Confidentiality, Integrity, Availability
EIS	Educational Information Systems
LMS	Learning Management Systems
SIS	Student Information Systems
DR	Digital Repositories
GIS	Geographic Information System
FAIR	Findable, Accessible, Interoperable, Reusable
AI	Artificial Intelligence
AP	Assessment Platform
HS	Harmonized Structure

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